

L2 SERIES

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Complete IT Infrastructure & System Administration Lab Guide

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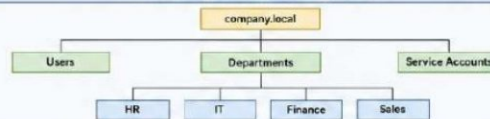
LAB 1: ACTIVE DIRECTORY USER MANAGEMENT

Objective: Create a new user, assign to OU, reset password, unlock account and disable terminated employee account.

LAB ENVIRONMENT

- Windows Server 2022
- Domain: company.local
- Domain Admin Credentials
- Tools: Active Directory Users and Computers (ADUC)

ORGANIZATIONAL UNIT STRUCTURE



1. CREATE A NEW USER

- Open Active Directory Users and Computers
- Right click on the OU (e.g. HR)
- Click New → User
- Enter user details:
 - First name: John
 - Last name: Smith
 - User login name: john.smith
 - User login name (UPN): john.smith@company.local
- Click Next
- Set Password: Welcome123!
- Uncheck "User must change password at next login"
- Click Next → Finish

2. ASSIGN TO DEPARTMENT OU

- In ADUC, navigate to company.local → Departments → HR
- The newly created user john.smith will appear under HR OU.
- This ensures proper organization and GPO application.

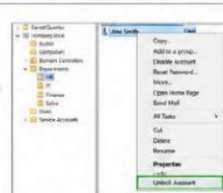


3. RESET PASSWORD

- Right click on the user (john.smith)
- Click Reset Password...
- Enter new password: NewPass123!
- Uncheck "User must change password at next login"
- Click OK
- Click OK to confirm

4. UNLOCK ACCOUNT

- Right click on the user (if account is locked)
- Click Unlock Account
- You will see a confirmation message.



5. DISABLE TERMINATED EMPLOYEE ACCOUNT

- Right click on the user (terminated employee)
- Click Disable Account
- The account will be disabled.
- The user will no longer be able to log in.



VERIFY ALL TASKS

- User John Smith created under HR OU
- Password reset successfully
- Account unlocked
- Terminated employee account disabled

SUMMARY OF COMMANDS (Optional PowerShell)

Create User <pre>New-ADUser -Name "John Smith" -GivenName "John" -Surname "Smith" -SamAccountName "john.smith" -UserPrincipalName "john.smith@company.local" -Path "OU=HR,OU=Departments,DC=company,DC=local" -AccountPassword (ConvertTo-SecureString "Welcome123!" -AsPlainText -Force) -Enabled \$true</pre>	Reset Password <pre>Set-ADAccountPassword -Identity "john.smith" -NewPassword (ConvertTo-SecureString "NewPass123!" -AsPlainText -Force) -Reset</pre>	Unlock Account <pre>Unlock-ADAccount -Identity "john.smith"</pre>	Disable Account <pre>Disable-ADAccount -Identity "john.smith"</pre>
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KEY POINTS

Always place users in correct OU • Use strong password policy • Unlock accounts instead of creating new one • Disable accounts instead of deleting • Follow least privilege principle

LAB 2: GROUP POLICY TROUBLESHOOTING

Scenario: Users are able to access Control Panel. The requirement is to block access to Control Panel for all users in the Sales OU.

LAB ENVIRONMENT

- Domain: company.local
- Domain Controller: DC01 (Windows Server 2022)
- Client: WIN10 (Windows 10/11)
- Tool: Group Policy Management (GPMC)
- User: sales.user (member of Sales OU)

OBJECTIVE

Prevent users in Sales OU from accessing Control Panel.

EXPECTED RESULT

Users in Sales OU will get the message:

STEP 1: CREATE AND CONFIGURE GPO

- Open Group Policy Management (gpmc.msc)
- Right click on Group Policy Objects → New
- Name it: Block_Control_Panel
- Right click the GPO → Edit
- Navigate to: User Configuration → Administrative Templates → Control Panel → Prohibit access to Control Panel
- Double click "Prohibit access to Control Panel"
- Select: Enabled
- Click Apply → OK

STEP 2: LINK GPO TO SALES OU

- In GPMC, right click Sales OU
- Click Link an Existing GPO...
- Select Block_Control_Panel
- Click OK
- The GPO is now linked to Sales OU

STEP 3: FORCE GROUP POLICY UPDATE

- Log on to client machine WIN10 with sales user
- Open Command Prompt as normal user
- Run the following command:
`gpupdate /force`
- You should see: "Computer Policy update has completed successfully." "User Policy update has completed successfully."

STEP 4: VERIFY RESULT

- Press Windows + R
- Type control
- Press Enter
- You should see the restriction message.

STEP 5: USE GPRESULT TO VERIFY

- Open Command Prompt
- Run:
`gpresult /r`
- Check the "Applied Group Policy Objects" section. You should see Block_Control_Panel listed.

STEP 6: TROUBLESHOOT INHERITANCE (IF NOT APPLIED)

- Right click Sales OU → Block Inheritance
- Select: Convert inherited policies into explicit policies on this object
- Click OK
- Run gpupdate /force again

TROUBLESHOOTING TIPS

- Make sure the user is in the Sales OU.
- Ensure the GPO is linked to the correct OU.
- Run gpresult /r to check which GPOs are applied.
- Check for GPO conflicts (Restricted Groups, Preferences, etc.).
- Ensure no other GPO is overriding this setting.
- Allow some time or run gpupdate /force.

VERIFICATION CHECKLIST

- ☒ GPO created and configured
- ☒ GPO linked to Sales OU
- ☒ gpupdate /force executed
- ☒ Control Panel access blocked
- ☒ gpresult /r shows GPO applied

KEY COMMANDS

Command	Purpose
<code>gpupdate /force</code>	Forces Group Policy update
<code>gpresult /r</code>	Shows result of applied GPOs
<code>gpresult /s report.html</code>	Generates HTML report
<code>rsop.msc</code>	Resultant Set of Policy (GUI tool)

LEARNINGS

- How to create and link a GPO
- How to configure Administrative Template settings
- How to force GPO update
- How to verify GPO application using gpresult
- How to troubleshoot inheritance issues

LAB 3: DNS ISSUE TROUBLESHOOTING

Scenario: Users are unable to access an internal application (app.company.local). You need to troubleshoot and fix the DNS issue.

1. LAB ENVIRONMENT

Domain: company.local
 DNS Server: DC01 (192.168.1.10)
 Client PC: WIN10 (192.168.1.50)
 Tool: DNS Manager, Command Prompt

NETWORK DIAGRAM

2. ISSUE

Users cannot access the internal application using the name **app.company.local**.

EXPECTED ERROR

3. TROUBLESHOOTING STEPS

1. Open Command Prompt on client PC.
2. Test the name resolution using nslookup.
3. Verify DNS server is reachable.
4. Check DNS service on DC01.
5. Verify DNS records for app.company.local.
6. If record is missing, create A record.
7. Flush DNS cache on client.
8. Test again to confirm resolution.

STEP-BY-STEP PROCEDURE

Step 1: Test name resolution

```

C:\> nslookup app.company.local
Server: DC01.company.local
Address: 192.168.1.10

*** DC01.company.local can't find
app.company.local: Non-existent domain
          
```

Step 2: Check DNS server reachability

```

C:\> ping 192.168.1.10

Pinging 192.168.1.10 with 32 bytes of data:
Reply from 192.168.1.10: bytes=32 time=1ms TTL=128
Reply from 192.168.1.10: bytes=32 time=1ms TTL=128
Reply from 192.168.1.10: bytes=32 time=1ms TTL=128
Reply from 192.168.1.10: bytes=32 time=1ms TTL=128

Ping statistics for 192.168.1.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
          
```

Step 3: Check DNS service on DC01

Step 4: Verify DNS records

Step 5: Create A record for app.company.local (if missing)

Step 6: Verify the new record

Step 7: Flush DNS cache on client

```

C:\> ipconfig /flushdns

Windows IP Configuration

Successfully flushed the DNS Resolver Cache.

C:\>
          
```

Step 8: Test again

```

C:\> nslookup app.company.local
Server: DC01.company.local
Address: 192.168.1.10

Name: app.company.local
Address: 192.168.1.20

C:\>
          
```

Step 9: Verify the application

VERIFICATION CHECKLIST

- ✓ nslookup initially failed
- ✓ DNS server reachable
- ✓ DNS service running on DC01
- ✓ A record created for app.company.local
- ✓ DNS cache flushed
- ✓ Name resolution successful

USEFUL COMMANDS

Command	Purpose
nslookup <name>	Test DNS name resolution
ping <ip>	Test connectivity to server
ipconfig /flushdns	Flush DNS cache
ipconfig /displaydns	View DNS cache
net stop dnscache	Stop DNS Client service
net start dnscache	Start DNS Client service

COMMON DNS RECORDS

Record Type	Purpose
A	Maps name to IPv4 address
AAAA	Maps name to IPv6 address
CNAME	Alias record (canonical name)
MX	Mail exchange record
NS	Name server record
PTR	Reverse lookup record

KEY POINTS

- Always verify DNS client settings.
- Check connectivity to DNS server.
- Ensure DNS service is running.
- Verify zone and records.
- Flush DNS cache after changes.
- Test again to confirm resolution.

LAB 4: DHCP FAILURE TROUBLESHOOTING

Scenario: New users are getting APIPA addresses (169.254.x.x). They cannot access network resources or internet.

1. LAB ENVIRONMENT

Domain: company.local
 DHCP Server: DC01 (192.168.1.10)
 Scope: 192.168.1.100 - 192.168.1.200
 Subnet Mask: 255.255.255.0
 Default Gateway: 192.168.1.1
 DNS Server: 192.168.1.10
 Client PC: WIN10 (DHCP Enabled)
 Tool: DHCP Manager, Command Prompt

NETWORK DIAGRAM

DC01 / DHCP Server
192.168.1.10

WIN10 / Client
(DHCP Enabled)

2. ISSUE

New users are getting APIPA addresses (169.254.x.x) instead of valid IP addresses from DHCP.

EXPECTED ERROR

```

Network Connection Details
Property      Value
IPv4 Address   169.254.10.25
Subnet Mask    255.255.0.0
Default Gateway
DHCP Server
DNS Servers
Lease Obtained
Lease Expires
  
```

3. TROUBLESHOOTING STEPS

1. Check client IP configuration.
2. Verify DHCP service is running on server.
3. Check DHCP scope for available addresses.
4. Check scope status and errors.
5. Renew IP address on client.
6. Release and renew IP address.
7. Expand DHCP scope (if needed).
8. Verify again IP configuration.
9. Test connectivity (ping gateway and DNS).

STEP 1: CHECK CLIENT IP CONFIGURATION

On Client PC (WIN10)

```

Command Prompt
C:\> ipconfig

Ethernet adapter Ethernet0:

   Connection-specific DNS Suffix . . : M0-dar1736622013M
   IPv4 Address. . . . . : 169.254.10.25
   Subnet Mask . . . . . : 255.255.0.0
   Default Gateway . . . . . :
  
```

* APIPA address detected (169.254.x.x)

STEP 2: CHECK DHCP SERVICE ON SERVER

On DC01 (DHCP Server)

```

Services (Local)

Name      Status      Startup Type
-----
DHCP Server      Running      Automatic
DNS Server       Running      Automatic
Windows Event Log Running      Automatic
TCP/IP NetBIOS Helper Running      Automatic
  
```

If not running, right click DHCP Server → Start.

STEP 3: CHECK DHCP SCOPE FOR AVAILABLE ADDRESSES

On DC01 - DHCP Manager

```

DHCP
  d01.company.local
    IPv4
      Scope [192.168.1.0] Main Scope
        Address Pool
        Reservations
        Scope Options
        Policies
        Server Options
        Policies
        Filters
        IPv6
  
```

Address Pool

Start IP Address	End IP Address	Total Addresses	Free Addresses
192.168.1.100	192.168.1.200	101	15

Ensure there are free addresses available.

STEP 4: CHECK SCOPE STATUS AND ERRORS

On DC01 - DHCP Manager

```

DHCP
  d01.company.local
    IPv4
      Scope [192.168.1.0] Main Scope
        Status: Active
        Start Time: 5/10/2024 9:00:00 AM
        End Time: Never
        Subnet Mask: 255.255.255.0
        Free IP Addresses: 15
        Percentage in Use: 85%
        Scope State: Active
  
```

Scope is Active and working properly.

STEP 5: RENEW IP ADDRESS ON CLIENT

On Client PC (WIN10)

```

Command Prompt
C:\> ipconfig /release
windows IP Configuration

No operation can be performed on Ethernet0
while it has its media disconnected.

C:\> ipconfig /renew
windows IP Configuration

An error occurred while renewing interface
Ethernet0 - Unable to contact your DHCP server.
  
```

STEP 6: RELEASE AND RENEW IP ADDRESS

On Client PC (WIN10)

```

Command Prompt
C:\> ipconfig /release
C:\> ipconfig /renew
windows IP Configuration

Ethernet adapter Ethernet0:

   IPv4 Address. . . . . : 192.168.1.120
   Subnet Mask . . . . . : 255.255.255.0
   Default Gateway . . . . . : 192.168.1.1
   DHCP Server . . . . . : 192.168.1.10
   DNS Servers . . . . . : 192.168.1.10
  
```

* Successfully obtained IP from DHCP

STEP 7: EXPAND DHCP SCOPE (IF NEEDED)

On DC01 - DHCP Manager

```

Address Pool
Start IP Address: 192.168.1.100
End IP Address: 192.168.1.200
  
```

To expand:

1. Right click Address Pool
2. Select Configure Options
3. Increase End IP Address (e.g., 192.168.1.250)
4. Click OK

STEP 8: VERIFY IP CONFIGURATION AGAIN

On Client PC (WIN10)

```

Command Prompt
C:\> ipconfig

Ethernet adapter Ethernet0:

   IPv4 Address. . . . . : 192.168.1.120
   Subnet Mask . . . . . : 255.255.255.0
   Default Gateway . . . . . : 192.168.1.1
   DHCP Server . . . . . : 192.168.1.10
   DNS Servers . . . . . : 192.168.1.10
   Lease Obtained . . . . . : 5/16/2024 10:15:22 AM
   Lease Expires . . . . . : 5/16/2024 2:15:22 PM
  
```

STEP 9: TEST CONNECTIVITY

On Client PC (WIN10)

```

C:\> ping 192.168.1.1
Reply from 192.168.1.1: bytes=32 time=1ms TTL=64
Reply from 192.168.1.1: bytes=32 time=1ms TTL=64
Reply from 192.168.1.1: bytes=32 time=1ms TTL=64
Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milliseconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
C:\> ping 192.168.1.10
Reply from 192.168.1.10: bytes=32 time=1ms TTL=128
* Connectivity successful
  
```

COMMON CAUSES AND FIXES

Issue	Fix
DHCP service stopped	Start DHCP Server service
No IP addresses in scope	Add or expand address pool
Scope inactive	Activate the scope
DHCP server not authorized	Authorize DHCP server in AD
Network connectivity issue	Check cables, switches, VLAN
Client firewall blocking DHCP	Allow DHCP (UDP: 67,68)
IP conflict	Check duplicates in address leases

VERIFICATION CHECKLIST

- ✓ Client gets IP from DHCP (not APIPA)
- ✓ DHCP service is running
- ✓ Scope has available addresses
- ✓ Client can ping gateway and DNS
- ✓ Internet access is working

USEFUL COMMANDS

Command	Purpose
ipconfig /all	View full IP configuration
ipconfig /release	Release current IP
ipconfig /renew	Renew IP from DHCP
ipconfig /flushdns	Flush DNS cache
ping <ip>	Test connectivity

DHCP PORTS (FOR REFERENCE)

Port	Protocol	Purpose
67	UDP	DHCP Server
68	UDP	DHCP Client
547	UDP	DHCPv6 Server
546	UDP	DHCPv6 Client

KEY POINTS

- APIPA (169.254.x.x) means client could not reach DHCP server.
- Always check DHCP service and scope before troubleshooting.
- Release and renew IP after making changes.
- Ensure clients can reach DHCP server on the network.
- Monitor DHCP logs in Event Viewer for errors.

LAB 5: FILE SERVER PERMISSION ISSUE

Scenario: Finance department users cannot access the shared folder \\FS01\Finance. Fix the share and NTFS permissions so only Finance users can access it.

1. LAB ENVIRONMENT

Domain : company.local
File Server : FS01 (192.168.1.10)
Client PC : WIN10 (192.168.1.50)
Share Name : Finance
Share Path : D:\Shares\Finance
Domain Group : Finance_Users
Test User : john.finance (member of Finance_Users)
Tool : File Explorer, Computer Management

NETWORK DIAGRAM

2. ISSUE

Users in Finance_Users group receive "Access is denied" when trying to access \\FS01\Finance.

EXPECTED ERROR

3. STEPS OVERVIEW

- Create shared folder on the server.
- Configure Share Permissions.
- Configure NTFS Permissions.
- Remove unnecessary permissions.
- Test access from client.
- Verify only Finance users can access.

4. CREATE SHARED FOLDER ON SERVER (FS01)

- On FS01, open File Explorer.
- Go to D:\Shares.
- Right click → New → Folder.
- Name it: Finance.

5. CONFIGURE SHARE PERMISSIONS

- Right click Finance folder → Properties.
- Go to sharing tab → Click share.
- Add group: Finance_Users.
- Permission Level: Change → Read/Write.
- Click Share → Done.

6. CONFIGURE NTFS PERMISSIONS

- Go to Security tab → Edit.
- Remove Users, Everyone and other groups.
- Click Add → Advanced.
- Find Name → Select Finance_Users → OK.
- Select Finance_Users and allow: Modify.
- Click Apply → OK.

7. REMOVE UNNECESSARY PERMISSIONS

- Select Users → Remove.
- Select Everyone → Remove.
- Keep only:
 - SYSTEM
 - Administrators
 - Finance_Users
- Ensure Finance_Users has Modify permission.

8. TEST ACCESS FROM CLIENT (WIN10)

- On WIN10, press Windows + R, type \\FS01\Finance and press Enter.
- Login with john.finance (member of Finance_Users).
- You should be able to open the folder and create files.

9. VERIFY RESTRICTIONS

- Login with any user NOT in Finance_Users.
- Try to access \\FS01\Finance.
- You should see:

VERIFICATION CHECKLIST

- Share \\FS01\Finance is created
- Share Permissions set for Finance_Users
- NTFS Permissions set for Finance_Users (Modify)
- Unnecessary permissions removed
- Users in Finance_Users can access
- Users not in group cannot access

USEFUL COMMANDS

Command	Purpose
\\FS01\Finance	Open the shared folder from client
icacls "D:\Shares\Finance"	View NTFS permissions
icacls "D:\Shares\Finance" /grant COMPANY\Finance_Users:(OI)CIJM	Grant Modify permission to group
net share	List all shares on the server
net share Finance /delete	Delete the share

SHARE PERMISSION LEVELS

Permission	Description
Full Control	Read, Write, Change & Delete
Change	Read, Write, Delete (no change permissions)
Read	View files and folders
Read/Write	View and write files
Special	Custom permissions

KEY POINTS

- Use Share Permissions to control access over the network.
- Use NTFS Permissions to secure the actual data.
- Always follow least privilege principle.
- Remove 'Users' and 'Everyone' for security.
- Test with both allowed and denied users.
- Document permission changes.

LAB 6: PRINT SERVER TROUBLESHOOTING

Issue: Users cannot print. The printer shows as "Offline". Fix the printer, clear the queue, reinstall the driver if needed and deploy via GPO.

1. LAB ENVIRONMENT

Domain : company.local
 Print Server : PS01 (192.168.1.20)
 Printer Name : HP LaserJet P2055
 Share Name : HP2055
 Client PC : WIN10 (192.168.1.50)
 User : john.finance (member of Finance_Users)
 Tool : Print Management, Services, GPO

NETWORK DIAGRAM

2. ISSUE

Users are unable to print. The printer shows as "Offline" on the client.

Devices and Printers (on client)

EXPECTED BEHAVIOR

Printer should show as "Ready" and documents should print successfully.

3. OVERVIEW OF STEPS

1. Check printer and server connectivity.
2. Check Print Spooler service.
3. Clear the print queue.
4. Reinstall or update the printer driver (if needed).
5. Test printing from client.
6. Deploy printer to users via GPO.
7. Verify from another client.

4. CHECK PRINTER CONNECTIVITY

On client PC, open Command Prompt and run:

```

C:\> ping 192.168.1.20

Pinging 192.168.1.20 with 32 bytes of data:
Reply from 192.168.1.20: bytes=32 time=1ms TTL=128
Reply from 192.168.1.20: bytes=32 time=1ms TTL=128
Reply from 192.168.1.20: bytes=32 time=1ms TTL=128
Reply from 192.168.1.20: bytes=32 time=1ms TTL=128

Ping statistics for 192.168.1.20:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milliseconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
C:\>

```

✓ If ping fails, fix network issues first.

5. CHECK PRINT SPOOLER SERVICE ON PS01

On PS01 server:

1. Press Win + R → type `services.msc` → Enter.
2. Find "Print Spooler".
3. Ensure Status is "Running" and Startup type is "Automatic".

Name	Status	Startup Type
Print Spooler	Running	Automatic
Server	Running	Automatic
Workstation	Running	Automatic

If not running:
 Right click → Start.
 If stopped frequently, check Event Viewer.

6. CLEAR PRINT QUEUE

1. On PS01, open Print Management.
2. Go to Print Management → Custom Filters → All Printers.
3. Right click on HP LaserJet P2055 → See what's printing.
4. Right click → Cancel All Documents.

7. REINSTALL / UPDATE PRINTER DRIVER (IF NEEDED)

1. On PS01, open Print Management.
2. Expand Print Servers → PS01 (local) → Drivers.
3. Right click HP LaserJet P2055 driver → Remove Driver Package.
4. Download latest driver from HP website.
5. Install driver: Right click Drivers → Add Driver → Choose the downloaded .inf file → Next.
6. Re-add the printer (if removed).

8. TEST PRINTING FROM CLIENT

1. On WIN10 client, open Devices and Printers.
2. Right click HP LaserJet P2055 → See what's printing (should be empty).
3. Click Printer → Print Test Page.
4. Printer should be "Ready" and print the page.

9. DEPLOY PRINTER VIA GPO

1. Open Group Policy Management (gpmc.msc).
2. Create or edit a GPO and link to Finance_Users OU.
3. Go to: User Configuration → Preferences → Control Panel Settings → Printers.
4. Right click Printers → New → Shared Printer.
5. Path: \\PS01\HP2055
6. Action: Update. Check "Reapply once then remove new" (optional).
7. Click OK and Close.

10. VERIFY FROM ANOTHER CLIENT

1. Log on to another client or run `gpupdate /force`
2. Check Devices and Printers.
3. HP LaserJet P2055 should appear automatically.
4. Print a test page and confirm.

11. USEFUL COMMANDS

Command	Purpose
<code>services.msc</code>	Open Services
<code>net stop spooler</code>	Stop Print Spooler service
<code>net start spooler</code>	Start Print Spooler service
<code>net stop spooler & del %systemroot%\System32\spool\PRINTERS*</code>	Clear print spool (stop service first)
<code>net start spooler</code>	Start Print Spooler service
<code>gpupdate /force</code>	Force Group Policy update
<code>runas /x:john.finance printui /s:PS01 /m:"HP LaserJet P2055" /q</code>	Add local printer via command line

12. COMMON CAUSES AND FIXES

Cause	Fix
Print Spooler is stopped	Start and set to Automatic
Documents stuck in queue	Clear print queue
Invalid or corrupt driver	Reinstall / update driver
Network issue	Check connectivity / firewall
Permission issue	Check share and NTFS permissions
Wrong port	Use correct port (Standard TCP/IP port)
GPO not applied	Run <code>gpupdate /force</code> and verify link

VERIFICATION CHECKLIST

- ✓ Printer is reachable (ping successful)
- ✓ Print Spooler service is running
- ✓ Print queue is clear
- ✓ Correct driver is installed
- ✓ Client can print test page
- ✓ Printer deployed via GPO
- ✓ Another client can print successfully

KEY POINTS

- Always check Spooler service first.
- Clearing queue resolves most issues.
- Use latest and correct driver.
- Test from multiple clients.
- GPO deployment saves time and ensures consistency.
- Check Event Viewer for detailed errors if issue persists.

LAB 6: PRINT SERVER TROUBLESHOOTING (PRINTER OFFLINE)

Scenario: Users are unable to print. Printer shows "Offline" on client machines. Fix the issue and make the printer available to all users.

1. LAB ENVIRONMENT

Print Server: PS01 (192.168.1.20)
Client PC: WIN10 (192.168.1.50)
Printer: HP LaserJet 1020
Share Name: HP1020
Port: IP: 192.168.1.30
Domain: company.local
Tool: Print Management, Services, Command Prompt

NETWORK DIAGRAM

2. ISSUE

Printer shows "Offline" on client machines and users cannot print.

EXPECTED RESULT

Printer should be online and users should be able to print.

3. TROUBLESHOOTING STEPS OVERVIEW

1. Check Print Spooler service on server.
2. Check printer is online on server.
3. Clear print queue (if any jobs stuck).
4. Verify printer driver.
5. Check printer port/IP connectivity.
6. Ensure share permission.
7. Test printing from server.
8. Test printing from client.

4. CHECK PRINT SPOOLER SERVICE

On PS01 (Server)

Name	Status	Startup Type
Print Spooler	Running	Automatic
Server	Running	Automatic
TCPIP NetBIOS Helper	Running	Automatic

If not running:
 ▶ Right click Print Spooler → Start
 ▶ Set Startup type → Automatic

5. CHECK PRINTER STATUS ON SERVER

Print Management

Printer Name: HP1020, Queue Status: Offline, Jobs in Queue: 2

If Offline:
 ▶ Right click HP1020 → Use Printer Online
 ▶ Right click → Properties → General → Ensure "Print to the following port" is correct.

6. CLEAR PRINT QUEUE (IF JOBS STUCK)

On PS01 (Server)

Print Management > Printers > HP1020

Document name: TestPage, Microsoft Word - Doc1, Report.pdf

Buttons: Cancel All Documents, Pause Printing, Resume Printing, Restart Spooler

Clear all documents and check again.

7. VERIFY PRINTER DRIVER

On PS01 (Server)

Devices and Printers

Right click HP LaserJet 1020 → Printer properties

Driver: HP LaserJet 1020

Status: Ready

No error shown

8. CHECK PRINTER PORT / IP CONNECTIVITY

On PS01 (Server)

Command Prompt

```

C:\> ping 192.168.1.30
Pinging 192.168.1.30: [bytes=32 time=1ms TTL=128]
Reply from 192.168.1.30: bytes=32 time=1ms TTL=128
Reply from 192.168.1.30: bytes=32 time=1ms TTL=128
Ping statistics for 192.168.1.30:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
  
```

If ping successful → Network is fine.

9. CHECK SHARE PERMISSION

On PS01 (Server)

HP1020 Properties

Sharing tab: HP1020 shared

Permissions tab: Full Control (Administrators), Read & execute (Authenticated Users), Change (Authenticated Users), Full control (Authenticated Users)

10. TEST PRINTING FROM SERVER

On PS01 (Server)

Test Page - Notepad

Print button

Date: 18/05/2024

If print successful → problem was on client.

11. TEST PRINTING FROM CLIENT

On WIN10 Client

HP1020 on PS01

Print button

Test Page - Notepad

Date: 18/05/2024

Print should be successful.

12. DEPLOY PRINTER TO ALL USERS (VIA GPO)

Group Policy Management

Policy: Default Domain Policy

Printers: Deployed Printers

In GPO: User Configuration > Control Panel Settings > Printers

HP1020 on PS01

13. COMMON ERROR MESSAGES & FIXES

Error	Fix
Printer Offline	Use Printer Online on server
Spooler not running	Start Print Spooler service
Driver not found	Reinstall correct printer driver
Access is denied	Check share & NTFS permissions
Port not found	Check IP and printer port
Print queue stuck	Clear all documents & restart spooler
Cannot connect	Check network, firewall, ping

14. PRINT SPOOLER COMMANDS (CMD)

```

C:\> net stop spooler
C:\> net start spooler
C:\> net stop spooler && net start spooler
C:\> del /q /f %systemroot%\system32\spool\PRINTERS\*.
  
```

Use above commands if queue is corrupted.

VERIFICATION CHECKLIST

- Print Spooler service running
- Printer online on server
- Print queue cleared
- Correct driver installed
- Port/IP connectivity verified
- Share permission configured
- Print successful from server
- Print successful from client

USEFUL COMMANDS

Command	Purpose
services.msc	Open Services
printmanagement.msc	Open Print Management
ping <ip>	Test printer connectivity
ipconfig /all	Check IP configuration
net stop spooler	Stop Print Spooler service
net start spooler	Start Print Spooler service

KEY POINTS

- Always check Print Spooler first.
- Ensure printer is online on the server.
- Clear stuck print jobs.
- Verify driver and port settings.
- Check share and NTFS permissions.
- Test from server before testing from client.
- Use GPO for deploying printers to users.

TROUBLESHOOTING FLOW

LAB 7: OUTLOOK ISSUE - OUTLOOK KEEPS ASKING FOR PASSWORD

Scenario: Outlook keeps asking for password repeatedly. Fix the issue by recreating profile, clearing credentials, checking Autodiscover and repairing OST.

1. LAB ENVIRONMENT

Domain: company.local
Mail System: Microsoft 365 (O365)
Client PC: WIN10 (192.168.1.50)
User: john.finance@company.com
Tool: Outlook, Control Panel, CMD, Credential Manager, Outlook ScanPST

NETWORK DIAGRAM

2. ISSUE

User opens Outlook. Outlook keeps asking for password.

EXPECTED BEHAVIOR

Outlook should connect without asking password.

3. OVERVIEW OF STEPS

1. Recreate Outlook profile.
2. Clear saved credentials.
3. Check Autodiscover.
4. Repair OST file.
5. Test Outlook connectivity.
6. Verify and confirm fix.

4. DETAILED STEPS TO RESOLVE THE ISSUE

1. Press Windows + R, type control mtzg32.cpl and press Enter.
2. Click Show Profiles.
3. Click Add, give profile name (e.g., Outlook).
4. Add email account and complete setup.
5. Set the new profile as "Always use this profile".
6. Click Apply and OK.

5. CLEAR SAVED CREDENTIALS

1. Open Control Panel -> Credential Manager.
2. Click Windows Credentials.
3. Find entries related to:
 - Outlook
 - MicrosoftOffice16_Data
 - company.com (or your domain)
4. Click Remove for each related entry.
5. Restart Outlook.

6. CHECK AUTODISCOVER

1. Open browser and go to: <https://outlook.office365.com/autodiscover/autodiscover.xml>
2. Sign in with your email and password.
3. You should see XML response.
4. If error, check DNS, Internet or proxy settings.

Autodiscover is working.

7. REPAIR OST FILE

1. Close Outlook.
2. Open ScanPST.exe (usually in: C:\Program Files\Microsoft Office\root\office16\).
3. Browse and select the OST file.
4. Click Start to scan.
5. Click Repair if issues are found.
6. Wait until repair is completed.
7. Restart Outlook.

8. TEST OUTLOOK CONNECTIVITY

1. Open Outlook.
2. Enter password once when prompted.
3. Outlook should open without asking password again.
4. Send/Receive test email.

Outlook is working correctly.

9. VERIFY AND CONFIRM FIX

- Outlook opens without asking password.
- Can send and receive emails.
- Calendar, Contacts and Tasks sync properly.
- No repeated password prompts.
- Issue resolved.

VERIFICATION CHECKLIST

- Outlook profile recreated successfully
- Old credentials removed
- Autodiscover working
- OST file repaired (if needed)
- Outlook opens without password prompt
- Email send/receive working
- Issue resolved and confirmed

USEFUL COMMANDS / PATHS

Command / Path	Purpose
control mtzg32.cpl	Open Mail (Profiles)
cmd /c rundll32.exe keymgr.dll, KRShowKeyMgr	Open Credential Manager
ping outlook.office365.com	Check connectivity
ipconfig /flushdns	Flush DNS cache
C:\Program Files\Microsoft Office\root\office16\SCANPST.EXE	Outlook Inbox Repair Tool

COMMON CAUSES AND FIXES

Cause	Fix
Corrupt Outlook profile	Recreate Outlook profile
Saved wrong credentials	Clear Credential Manager
Autodiscover not working	Check URL/DNS/Proxy
Corrupt OST file	Run ScanPST and repair
Network / MFA issues	Check Internet and MFA

KEY POINTS

- Always start with profile recreation.
- Remove old/incorrect credentials.
- Verify Autodiscover in browser.
- Repair OST if Outlook data file is corrupted.
- Test email functionality after each step.
- Document the actions and resolution.

LAB 8: OFFICE 365 USER ADMINISTRATION

Scenario: Perform day-to-day user administration tasks in Microsoft 365 admin center.

1. LAB ENVIRONMENT

Tenant Name : company.onmicrosoft.com
 Admin Account : admin@company.onmicrosoft.com
 Platform : Microsoft 365 Admin Center
 Client PC : WIN10 (192.168.1.50)
 Tool : Microsoft 365 admin center
 URL : <https://admin.microsoft.com>

NETWORK DIAGRAM

2. CREATE USER WITH MAILBOX

- Go to Microsoft 365 admin center.
- Users > Active users > Add a user.
- Add name, username, domain and password.
- Assign product license.
- Click Add user.

Add a user

First name: John, Last name: Finance
 Display name: John Finance
 Username: john.finance@company.onmicrosoft.com
 Password: [Generated]
☒ Automatically make a password
☒ Require this user to change their password when they first sign in

Result: User john.finance@company.onmicrosoft.com created.

3. ASSIGN LICENSE

- In Active users, select the user.
- Click Licenses and apps.
- Select a license (e.g., Microsoft 365 Business Standard).
- Click Save changes.

Licenses and apps

Select location: United Arab Emirates
 License: Microsoft 365 Business Standard (1 of 25 licenses available)
 Assign: [X] X

Result: License assigned successfully.

4. CREATE SHARED MAILBOX

- Go to Teams & groups > Shared mailboxes.
- Click Add a shared mailbox.
- Enter name and email address.
- Add owners and members.
- Click Create.

Add a shared Mailbox

Name: Finance Shared Mailbox
 Email: finshared@company.onmicrosoft.com
 Add owners: John Finance [X] X
 Add members: Jane Accounts [X] X

Result: Shared mailbox finshared@company.onmicrosoft.com created.

5. CREATE DISTRIBUTION GROUP

- Go to Teams & groups > Groups.
- Click Add a group > Distribution list.
- Enter name and description.
- Add members.
- Click Create.

Add a distribution list

Group name: All Employees
 Group email address: allem@company.onmicrosoft.com
 Group description: Group for all employees
 Add members: John Finance [X] X, Jane Accounts [X] X

Result: Distribution group created successfully.

6. PASSWORD RESET

- Go to Active users.
- Select the user.
- Click Reset password.
- Choose to Auto-generate password or enter a new password.
- Check "Require this user to change password when they next sign in".
- Click Reset password.

Reset password

User: John Finance
☒ Auto-generate password
☐ Let me create the password
☒ Require this user to change their password when they next sign in

Result: Password reset successfully.

7. ADD USER TO GROUP

- Go to Teams & groups > Groups.
- Select the group (e.g., All Employees).
- Click Edit.
- Add members.
- Click Save.

All Employees

Members (5): John Finance, Jane Accounts, John Smith, Jane Smith, John Doe

Result: User added to the group.

8. MODIFY USER PROPERTIES

- Go to Active users.
- Select the user.
- Click Manage username, profile or roles.
- Update properties (e.g., display name, phone).
- Click Save changes.

John Finance

Display name: John Finance
 Phone number: +971 501234567
 Job title: Finance Executive

Result: User properties updated.

9. VERIFY MAILBOX AND SIGN-IN

- Sign in as the user at <https://outlook.office.com>
- Verify mailbox access.
- Send and receive a test email.

Outlook

Home View Help
 New mail
 Favorites: Inbox, Sent items, Drafts
 Inbox: Jane Accounts sent email This is a test email. 10:15 AM

Result: Mailbox access verified successfully.

10. VERIFY LICENSE ASSIGNMENT

- Go to Active users.
- Select the user.
- Under Licenses and apps, verify the license status is "Active".

Licenses and apps

Microsoft 365 Business Standard: Active
 27 of 25 licenses available

Result: License is active for the user.

VERIFICATION CHECKLIST

- ✓ User with mailbox created
- ✓ License assigned and active
- ✓ Shared mailbox created
- ✓ Distribution group created
- ✓ Password reset successfully
- ✓ User added to group
- ✓ Mailbox login and email test successful

USEFUL ADMIN CENTER PATHS

Task	Path
Active users	Users > Active users
Add user	Users > Active users > Add a user
Licenses	Users > Active users > Licenses and apps
Shared mailbox	Teams & groups > Shared mailboxes
Groups	Teams & groups > Groups
Reset password	Users > Active users > Reset password

COMMON CAUSES AND FIXES

Issue	Fix
License not assigned	Assign a valid license
Cannot sign in	Check password and license
Mailbox not found	Wait a few minutes and retry
Group not showing	Refresh or check membership
Email not received	Check spam/junk and send again

KEY POINTS

- Always assign a license to enable service.
- Use strong passwords and enforce change at first sign-in.
- Use shared mailboxes for team access.
- Use distribution groups to send emails to multiple users.
- Changes may take a few minutes to reflect.

LAB 9: OUTLOOK EMAIL AND SYNC ISSUES - TROUBLESHOOTING

Scenario: Outlook is not sending or receiving emails, folders are not syncing and search is not working. Diagnose and fix the issues.

1. LAB ENVIRONMENT

User: john@franciscocompany.com
Domain: company.local
Mail System: Microsoft 365 (O365)
Client PC: WIN10 (192.168.1.50)
Tool: Outlook, Test Email, Connectivity Analyzer, Safe Mode, OST Scan/PST

NETWORK DIAGRAM

2. ISSUE

User reports the following:

- Not sending or receiving emails
- Folders not syncing
- Search is not returning results

EXPECTED BEHAVIOR

Outlook should send/receive emails, sync all folders and search should return results.

3. OVERVIEW OF STEPS

- Check internet and Outlook connectivity.
- Check Outlook Send/Receive settings.
- Clear Outlook cache.
- Reset Send/Receive groups.
- Update Outlook.
- Check mailbox size and quotas.
- Test in Outlook Safe Mode.
- Repair Outlook profile.
- Scan and repair OST file.
- Verify and confirm fix.

4. CHECK CONNECTIVITY

- Close Outlook → File → Account Settings → Account Settings.
- Ensure account status is "Online".
- Click "Test Account Settings".

Connectivity is working.

5. CHECK SEND/RECEIVE SETTINGS

- Go to Send / Receive tab → Send/Receive Groups → Define Send/Receive Groups.
- Ensure all accounts are checked.
- Set "Schedule an automatic send/receive every 15 minutes".
- Click Close → Send/Receive All Folders.

6. CLEAR OUTLOOK CACHE

- Close Outlook.
- Press Windows + R, type: %localappdata%\Microsoft\Outlook
- Delete all files in this folder.
- Reopen Outlook.

Cache cleared.

7. RESET SEND/RECEIVE GROUPS

- Send/Receive tab → Send/Receive Groups → Define Send/Receive Groups.
- Select the group and click Remove.
- Click Now, add all accounts.
- Click OK → Send/Receive All Folders.

Group removed and recreated.

8. UPDATE OUTLOOK

- File → Office Account → Update Options → Update Now.
- Restart Outlook.

Outlook is up to date.

9. CHECK MAILBOX SIZE AND QUOTAS

- Go to FILE → FOLDER > FOLDERS > FOLDERS → Library → Active Users → select user.
- Check Mailbox storage.
- Ensure storage is not near quota.

Storage is normal.

10. TEST IN SAFE MODE

- Close Outlook.
- Press Windows + R, type: outlook.exe /safe
- Test send/receive and search.

Outlook works in Safe Mode.

11. REPAIR OUTLOOK PROFILE

- Control Panel → Mail → Show Profiles.
- Select profile → Properties.
- Click Email Accounts → Repair.
- Test Outlook.

Profile repaired.

12. SCAN AND REPAIR OST FILE

- Close Outlook.
- Press Windows + R, type: scandpst.exe
- Browse to the OST file location (the default path shown below).
- Click Start.
- If errors found, click Repair.

OST scan and repair completed.

13. VERIFY AND CONFIRM FIX

- Open Outlook normally.
- Click Send/Receive All Folders (F9).
- Check Inbox for new emails.
- Sync folders.
- Use search to find emails.

Emails received, folders synced, search working.

14. TEST FROM ANOTHER CLIENT

- Open Outlook on another device.
- Login with same account.
- Verify emails and folders.
- Send a test email.
- Confirm email sent and received.

Works from another client.

15. DOCUMENT AND CLOSE

- Document the root cause and steps taken.
- Update ticket.
- Inform the user.
- Close the incident.

Issue resolved and documented.

VERIFICATION CHECKLIST

- Internet and Outlook connectivity verified
- Send/Receive settings configured
- Outlook cache cleared
- Send/Receive groups reset
- Outlook updated
- Mailbox quota checked
- Works in Safe Mode
- Profile repaired
- OST File repaired (if needed)
- Issue resolved and confirmed

USEFUL COMMANDS / PATHS

Command / Path	Purpose
outlook.exe /safe	Run Outlook in Safe Mode
scandpst.exe	Scan and repair OST file
%localappdata%\Microsoft\Outlook	Outlook cache location
Win + R	Open Run dialog
Send/Receive All Folders (F9)	Send/Receive emails

COMMON CAUSES AND FIXES

Cause	Fix
Connectivity issue	Check internet / account settings
Send/Receive misconfigured	Reset Send/Receive groups
Outdated Outlook	Update Outlook
Cache corruption	Clear Outlook cache
Quota or OST file	Scan and repair OST file
Profile issue	Repair Outlook profile or recreate

KEY POINTS

- Always check connectivity first.
- Reset Send/Receive groups if emails are stuck.
- Clear cache to fix sync and search issues.
- Use Safe Mode to identify add-in or profile issues.
- Repair OST file if folders are not syncing.
- Verify from another client before closing the ticket.

LAB 10: ONEDRIVE AND SHAREPOINT - COLLABORATION AND PERMISSIONS

Scenario: Configure OneDrive and SharePoint, create sites and libraries, sync files, share with users and manage permissions.

1. LAB ENVIRONMENT

Tenant : company.onmicrosoft.com
Admin Account : admin@company.onmicrosoft.com
User 1 : john.finance@company.com
User 2 : jane.accounts@company.com
Platform : Microsoft 365
Client PC : WIN10 (192.168.1.50)
Tool : OneDrive, SharePoint Admin Center, SharePoint, Teams (optional)
URL : https://admin.microsoft.com

NETWORK DIAGRAM

WIN10 Client
192.168.1.50

SharePoint
OneDrive

2. CREATE ONEDRIVE

1. Sign in to https://portal.office.com
2. Click the app launcher (9 dots)
3. Click OneDrive.
4. Click New > Folder and name it "Project Files".
5. Upload a file and verify.

OneDrive created and file uploaded.

3. CREATE SHAREPOINT SITE

1. Go to https://admin.microsoft.com
2. Admin centers > SharePoint
3. Click Active sites > Create site
4. Choose "Team site"
5. Site name: Finance Team Site
6. Add members: john.finance@company.com, jane.accounts@company.com
7. Click Finish.

SharePoint site created.

4. CREATE DOCUMENT LIBRARY

1. Open the site in SharePoint.
2. Click Settings > Site contents.
3. Click New > Document library.
4. Name: "Finance Documents"
5. Click Create.

Document library created.

5. SYNC DOCUMENT LIBRARY

1. In the library, click Sync.
2. OneDrive will open and sync the library.
3. Files will be available offline.
4. Add or modify a file and verify sync.

Library synced with OneDrive.

6. SHARE FILE / FOLDER

1. In the library, select a file or folder.
2. Click Share.
3. Enter user: jane.accounts@company.com
4. Choose permission: Can edit.
5. Add a message and click Send.

File shared successfully.

7. MANAGE PERMISSIONS

1. Click -- (---) > Manage access.
2. View existing permissions.
3. Click Advanced > Stop inheriting permissions.
4. Remove unnecessary permissions if any.
5. Click Save.

Permissions updated.

8. CREATE SHAREPOINT LIST

1. On the site, click New > List.
2. Name: "Project Tracker"
3. Choose Blank list.
4. Add columns: Task (Single line), Owner (Person), Status (Choice), Due Date (Date)
5. Add sample items.

List created and items added.

9. AUTOMATE WITH POWER AUTOMATE (OPTIONAL)

1. Go to Power Automate (make.powerautomate.com)
2. Create > Automated cloud flow.
3. Trigger: When an item is created (SharePoint).
4. Action: Send an email (V2).
5. Save and test the flow.

Flow created and tested.

10. TEST COLLABORATION

1. Login as User 2 (jane.accounts@company.com).
2. Open the shared file/folder.
3. Edit and save the file.
4. Verify changes reflect for User 1.

Collaboration verified.

VERIFICATION CHECKLIST

- OneDrive folder created
- Files uploaded and accessible
- SharePoint site created
- Document library created
- Library synced with OneDrive
- File/folder shared with a user
- Permissions configured correctly
- SharePoint list created
- Collaboration tested successfully

USEFUL COMMANDS / URLS

Command / URL	Purpose
https://portal.office.com	Access Microsoft 365 apps
https://admin.microsoft.com	Access Microsoft 365 admin center
https://tenantid.sharepoint.com	Access SharePoint sites
PowerShell: Connect-SPService -URI https://tenantid-admin.sharepoint.com	Connect to SharePoint Online
OneDrive Sync Client	Sync libraries to local PC

COMMON CAUSES AND FIXES

Issue	Fix
Cannot find shared item	Check sharing link and permissions
Sync not working	Check OneDrive status and sign-in
Access denied	Verify user permissions
Changes not visible	Refresh page / wait and retry
Library not syncing	Check folder path and quota
List not editable	Check list permissions

KEY POINTS

- Use OneDrive for personal files and SharePoint for team files.
- Always share with the least privilege (Can view / Can edit).
- Sync libraries to work offline and improve productivity.
- Manage permissions carefully to protect data.
- Lists help track and manage team information.
- Automate notifications and tasks with Power Automate.

LAB 11: MICROSOFT TEAMS AND PLANNER – TEAM COLLABORATION AND TASK MANAGEMENT

Scenario: Create a Team, add members, have a channel conversation, schedule a meeting and create a plan in Planner, assign tasks and track progress.

1. LAB ENVIRONMENT

Prereq: <https://www.officeapps.microsoft.com/onenote.aspx?ui=en&rs=en&ad=us>
Admin Account: admin@company.onmicrosoft.com
User 1: john.financial@company.com
User 2: jane.accounts@company.com
Platform: Microsoft 365
Client PC: WIN10 (192.168.1.50)
Tool: Microsoft Teams, Planner
URL: <https://teams.microsoft.com>

NETWORK DIAGRAM

2. CREATE A TEAM

- Go to <https://teams.microsoft.com> and sign in.
- Click Teams (left menu) > Join or create a team.
- Select Create a team.
- Choose From scratch.
- Enter team name: Finance Team.
- Add description (optional).
- Click Create.

Create your team

From scratch (You'll only see this team)

From a private team (Create your team from an Microsoft 365 group that you own)

From an existing team (Add to a team you've already created)

Team "Finance Team" created.

3. ADD MEMBERS TO TEAM

- Open the Finance Team.
- Click >> (More options) next to team name > Manage team.
- Go to Members tab.
- Click Add member.
- Add members: john.financial@company.com, jane.accounts@company.com
- Click Add.

Members added successfully.

4. CREATE A CHANNEL & START CONVERSATION

- In Finance Team, click Channels > Add channel.
- Channel name: General.
- Keep Standard - Everyone on the team has access.
- Click Add.
- Click General channel and start a conversation.
- Type a message and click Send.

Conversation started in General channel.

5. SCHEDULE A MEETING

- Click Calendar (left menu) > New meeting.
- Title: Budget Review Meeting.
- Add attendees: team members.
- Set date and time.
- Click Send.

Meeting scheduled.

6. ADD TABS (FILES & PLANNER)

- In General channel, click >> (Add a tab).
- Add tabs: Files and Planner.
- Click Save.

Files and Planner tabs added.

7. CREATE A PLAN IN PLANNER

- Click the Planner tab.
- Click New plan.
- Plan name: Q2 Budget Plan.
- Add description (optional).
- Choose to create a new plan.
- Click Create plan.

Plan "Q2 Budget Plan" created.

8. ADD BUCKETS

- Open Q2 Budget Plan.
- Click >> Add new bucket.
- Add buckets:
 - Planning
 - Execution
 - Review
 - Completed
- Buckets created.

Buckets added.

9. CREATE AND ASSIGN TASKS

- In Planning bucket, click >> Add task.
- Task 1: Collect department budgets. Assign to: john.financial@company.com
- Task 2: Consolidate budget data. Assign to: jane.accounts@company.com
- Click Add task for each.
- Add more tasks as needed.

Tasks created and assigned.

10. UPDATE TASK PROGRESS

- Open a task.
- Click Set progress and choose:
 - Not started
 - In progress
 - Completed
- Move task to next bucket by drag & drop.

Progress updated and tasks moved.

11. VIEW PLAN PROGRESS

- In Planner, click Charts tab.
- View status of tasks by bucket and members.
- Click Board to return.

Plan progress reviewed.

12. FILE SHARING IN TEAM

- Go to Files tab in General channel.
- Click Upload > Files.
- Upload a file (e.g., Q2_Budget.xlsx).
- File available to all team members.

File uploaded and shared.

13. CHAT IN TEAM

- Click Chat (left menu).
- Start a new chat with team members.
- Send a message.

Chat completed.

14. VERIFY ON MOBILE (OPTIONAL)

- Open Microsoft Teams mobile app.
- Sign in with user account.
- Verify team, channels, files and tasks.

Team visible on mobile.

15. CLEAN UP (OPTIONAL)

- Delete plan if no longer needed.
- Delete team if created for lab only.
- Remove members (if required).
- Sign out.

Cleanup completed.

VERIFICATION CHECKLIST

- Team created successfully
- Members added to team
- Channel conversation started
- Meeting scheduled
- Plan created in Planner
- Buckets created
- Tasks created and assigned
- Task progress updated
- Files shared in team
- Chat completed
- Plan progress reviewed

USEFUL COMMANDS / URLS

Command / URL: <https://teams.microsoft.com>
Access Planner: <https://planner.officeapps.microsoft.com>
Start a new chat: @John (Teams search)
Quick command in search: @jane (Teams search)
Remove team members: @jane (Teams search)

COMMON CAUSES AND FIXES

Issue	Fix
Cannot create team	Check permissions and license
Members not added	Ensure correct email and license
Planner tab not visible	Click >> to add Planner tab
Tasks not assigned	Assign task to a member
Changes not visible	Refresh or check internet

KEY POINTS

- Teams is for communication, meetings and file sharing.
- Planner is for task management and tracking.
- Use channels to keep conversations organized.
- Assign tasks clearly and update progress regularly.
- Collaborate, communicate and track to deliver on time!

LAB 12: POWER AUTOMATE – AUTOMATE TASKS AND APPROVALS

Scenario: Create flows in Power Automate to automate tasks, send approvals and notifications.

1. LAB ENVIRONMENT

Browser
 1. companynamecloudsoft.com
User
 1. john.finance@company.com
Platform
 1. Power Automate (make.powerautomate.com)
License
 1. Microsoft 365 E3
Client PC
 1. WIN10 (192.168.1.58)
Tool
 1. Power Automate (Web)

NETWORK DIAGRAM

2. SIGN IN TO POWER AUTOMATE

- Go to <https://make.powerautomate.com>.
- Sign in with your account.
- You will land on Power Automate home.

Signed in successfully.

3. CREATE AUTOMATED CLOUD FLOW

- Click Create > Automated cloud flow.
- Flow name: New File Notification.
- Choose trigger: When a file is created (OneDrive for Business).
- Click Create.

Automated flow created.

4. CONFIGURE FLOW - NEW FILE NOTIFICATION

- Trigger is set. When a file is created (properties only).
- Click + New step.
- Search and select: Send an email (V2).
- Configure:
 - To: john.finance@company.com
 - Subject: New file created
 - Body: A new file has been created.
- Click Save.

Flow configured and saved.

5. TEST NEW FILE NOTIFICATION FLOW

- Upload a file to OneDrive.
- Flow should trigger automatically.
- Check email for notification.

Notification email received.

6. CREATE APPROVAL FLOW

- Click Create > Automated cloud flow.
- Flow name: Purchase Request Approval.
- Trigger: When a new item is created (SharePoint).
- Click Create.

Approval flow created.

7. CONFIGURE APPROVAL FLOW

- Click + New step.
- Search and select: Start and wait for an approval.
- Configure:
 - Approval type: Approval/Reject - First to respond
 - Title: New Purchase Request
 - Assigned to: john.finance@company.com
 - Details: Please review the purchase request.
- Click + New step > Condition.
- Set condition: Outcome is equal to Approve.
- Add actions:
 - If Yes: Send an email (Approved)
 - If No: Send an email (Rejected)
- Click Save.

Approval flow configured.

8. TEST APPROVAL FLOW

- Go to the SharePoint list (e.g., Purchase Requests).
- Add a new item.
- Approver (Jane Account) receives approval request.
- Approver selects Approve.
- Requester receives approval email.

Approval flow tested successfully.

9. CREATE SCHEDULED FLOW

- Click Create > Scheduled cloud flow.
- Flow name: Daily Report.
- Set recurrence: Everyday, 8:00 AM.
- Click Create.

Scheduled flow created.

10. CONFIGURE AND TEST SCHEDULED FLOW

- Click + New step.
- Search and select: Send an email (V2).
- To: team@company.com
- Subject: Daily Report
- Body: Please find attached the daily report.
- Click Save.
- Flow runs daily at 8:00 AM and sends email.

Scheduled flow configured and tested.

11. MANAGE FLOWS

- Go to My flows.
- You can:
 - Edit
 - Turn on/off
 - View run history
 - Delete

Flows managed.

12. SOLUTIONS (OPTIONAL)

- Go to Solutions > New solution.
- Create a solution: Finance Automation.
- Add flows to the solution.
- Save.

Solution created and flows added.

VERIFICATION CHECKLIST

- Automated flow sends email on new file
- Approval flow sends request and handles response
- Approval emails (Approved/Rejected) received
- Scheduled flow sends daily email
- All flows are active and working
- Flows visible in My flows
- Solution created (optional)

UMBRAL COMMAND / URL	Purpose	Issue	Fix
https://make.powerautomate.com	Power Automate portal	Flow not triggering	Check trigger setup and connections
Create > Automated cloud flow	Creates new request flow	Approval not received	Check assigned to and approval email
Create > Scheduled cloud flow	Creates time-based flow	Email not sent	Check the address and email history
Start and wait for an approval	Add approval step	Flow run failed	Check run history and fix errors
My flows	View and manage flows	Permission error	Reassign and verify permissions
Solutions	Organize flows in solutions	Schedule not running	Check time zone and schedule status

KEY POINTS

- Use flows to save time and reduce manual work.
- Approvals help in decision-based processes.
- Triggers start flows automatically.
- Test flows to ensure they work as expected.
- Monitor run history to troubleshoot issues.
- Use solutions to organize and share flows.

LAB 13: MICROSOFT PURVIEW DLP – CREATE POLICIES AND TEST

Scenario: Create a DLP policy to protect sensitive information, apply it to locations, test the policy and review alerts.

1. LAB ENVIRONMENT

Browser: companyonmicrosoft.com
User: john.finance@company.com
Platform: Microsoft Purview compliance portal
License: Microsoft 365 E3
Client PC: WIN10 (192.168.1.50)
Tool: Microsoft Purview
URL: https://compliance.microsoft.com

NETWORK DIAGRAM

WIN10 Client
192.168.1.50

Microsoft 365

Microsoft Purview Compliance Portal

2. SIGN IN TO MICROSOFT PURVIEW

- Go to <https://compliance.microsoft.com>
- Sign in with your account.
- You will land on the Microsoft Purview compliance portal.

Home

Welcome to the Microsoft Purview compliance portal.

Use the navigation to manage compliance solutions for your organization.

Home

Compliance Manager

Data classification

Data connection

Alerts

Policy & groups

Solutions

Settings

Signed in successfully.

3. CREATE DLP POLICY

- Go to Solutions > Data loss prevention.
- Click + Create policy.
- Choose a template: Financial.
- Click Next.

Solutions > Data loss prevention

Overview Policies Alerts Endpoint DLP

Create policy Refresh

Name: Negative brand

Template "Financial" selected.

4. NAME AND ASSIGN POLICY

- Name: Protect Financial Data
- Description: Protect sensitive financial information.
- Click Next.
- Choose locations to apply this policy later:

Name your policy

Name: Protect Financial Data

Description: Protect sensitive financial information.

Back Next

Policy named and description added.

5. DEFINE POLICY RULE

- Click + Add rule.
- Name: Block Financial Data.
- Condition: Content contains listed sensitive information types.
- Add: Credit Card Number, Bank Account Number.
- Action: Block the content.
- Click Done.

Define policy rule

Click + Add rule.

Name: Block Financial Data.

Condition: Content contains listed sensitive information types.

Add: Credit Card Number, Bank Account Number.

Action: Block the content.

Click Done.

Rule defined.

6. SET POLICY MODE

- Choose policy mode.
- Start with: Test with notifications.
- Click Next.

Set a policy mode

Enforce: Enforce the policy and apply the actions.

Test with notifications: Test the policy and send notifications when a match occurs.

Test without notifications: Test the policy without sending notifications.

Back Next

Mode set to "Test with notifications".

7. REVIEW AND CREATE POLICY

- Review the settings.
- Click Submit.
- Policy status will be "On".

Review your settings

Policy name: Protect Financial Data

Mode: Test with notifications

Locations: SharePoint sites, OneDrive accounts, Exchange email

Rules: 1 rule

Last modified: 5/23/2024 12:30 AM

Back Submit

Policy created and turned on.

8. APPLY POLICY TO LOCATIONS

- Go to the policy > Edit locations.
- Select the locations.
- SharePoint sites
- OneDrive accounts
- Exchange email
- Click Save.

Edit locations

SharePoint sites

OneDrive accounts

Exchange email

Save Cancel

Locations selected and saved.

9. TEST POLICY - SHAREPOINT

- Go to a SharePoint site > Documents.
- Create a file "TestData.txt".
- Add sample credit card number: 4111 1111 1111 1111
- Save and close.
- Upload the file to the Documents library.

SharePoint > Documents

Create a file "TestData.txt".

Add sample credit card number: 4111 1111 1111 1111

Save and close.

Upload the file to the Documents library.

File uploaded with sensitive data.

10. TEST POLICY - ONEDRIVE

- In OneDrive, create a file "TestData.txt".
- Add sample bank account number: 123456789012
- Save the file.
- Wait a few minutes for DLP to scan.

OneDrive > My files

Create a file "TestData.txt".

Add sample bank account number: 123456789012

Save the file.

Wait a few minutes for DLP to scan.

File saved in OneDrive.

11. TEST POLICY - EXCHANGE EMAIL

- Compose a new email.
- Add the credit card number in the body.
- Send the email to a test user.
- Wait a few minutes for DLP to scan.

Compose a new email

Add the credit card number in the body.

Send the email to a test user.

Wait a few minutes for DLP to scan.

Test email sent.

12. CHECK ALERTS

- Go to Solutions > Data loss prevention.
- Click Alerts
- Review the alerts generated by the policy.

Solutions > Data loss prevention

Overview Policies Alerts Endpoint DLP

Alerts

Alerts generated by the policy.

13. VIEW ALERT DETAILS

- Click on alert to view details.
- Review matched content and activity.
- Close the details pane.

Alert details

Policy name: Protect Financial Data

Rule: Block Financial Data

Location: SharePoint

Activity: File uploaded

Account: john.finance@company.com

Date: 5/23/2024 10:45 AM

Matched content: 4111 1111 1111 1111

Close

Alert details reviewed.

14. TURN POLICY TO ENFORCE (OPTIONAL)

- Go to the policy.
- Click Edit policy settings.
- Change mode to Enforce.
- Click Save.

Edit policy settings

Policy mode: Enforce

Test with notifications: Test the policy and send notifications.

Test without notifications: Test the policy without sending notifications.

Save Cancel

Policy mode changed to Enforce.

15. SUMMARY

DLP policy created successfully!

- Policy created from template.
- Rule defined to protect financial data.
- Applied to SharePoint, OneDrive and Exchange.
- Tested and alerts received.
- Policy can now be enforced.

Summary

DLP policy created successfully!

- Policy created from template.
- Rule defined to protect financial data.
- Applied to SharePoint, OneDrive and Exchange.
- Tested and alerts received.
- Policy can now be enforced.

VERIFICATION CHECKLIST

- Signed in to Microsoft Purview
- Created DLP policy
- Defined rule for sensitive info
- Selected policy mode
- Applied policy to locations
- Tested in SharePoint
- Tested in OneDrive
- Tested in Exchange email
- Checked alerts
- Reviewed alert details
- (Optional) Enforced the policy

USEFUL COMMANDS / URLS

Command / URL	Purpose
https://compliance.microsoft.com	Open Microsoft Purview portal
Solutions > Data loss prevention	Open DLP solution
+ Create policy	Create a new DLP policy
Alerts	View DLP alerts
Edit locations	Apply policy to locations
Policy mode: Enforce	Enforce the policy

COMMON CAUSES AND FIXES

Issue	Fix
No alerts generated	Test for scan or check scope and rule
Items not scanned	Ensure location is in policy scope
Policy not working	Check mode and rule configuration
False positives	Refine rules and sensitive info types

KEY POINTS

- DLP protects sensitive information.
- Start with Test mode to validate rules.
- Apply policy to the right locations.
- Use alerts to monitor and investigate.
- Enforce policy after successful testing.

Note: DLP scans run periodically. It may take 5-15 minutes (or more) for alerts to appear.

LAB 14: SCCM SOFTWARE DEPLOYMENT

Scenario: Package and deploy 7-Zip to Windows 10 clients using SCCM and monitor the deployment.

1. LAB ENVIRONMENT

Site Code : MEM
 SCCM Server : MEMCM (192.168.1.10)
 Primary Site : MEM - MEMCM Primary Site
 Console : Configuration Manager Console
 Client OS : Windows 10
 Domain : company.com
 Tool : Microsoft SCCM ConfigMgr
 Purpose : Deploy 7-Zip to Windows 10 clients

NETWORK DIAGRAM

2. DOWNLOAD 7-ZIP INSTALLER

- Go to <https://www.7-zip.org/>
- Download 7-Zip (x64) installer.
- Save the file to a local folder.

7-Zip is a file archiver with a high compression ratio.

Link	Type	Windows	Size
Download	x64	64-bit x64	1.1 MB
Download	x86	32-bit x86	1.1 MB

License

Installer downloaded.

3. CREATE APPLICATION

- In SCCM Console, go to Software Library > Overview > Application
- Click Create Application
- Select Manually specify the application information.
- Click Next.

Create Application Wizard

Specify settings for this application

General

Application Name: 7-Zip File Archiver

Administrator comments: 7-Zip File Archiver

Author: Igor Pavlov

Software version: 23.01

Application creation started.

4. CONFIGURE APPLICATION DETAILS

Application Information

Name: 7-Zip

Administrator comments: 7-Zip File Archiver

Author: Igor Pavlov

Software version: 23.01

Application details configured.

5. ADD DEPLOYMENT TYPE

- Click Add > New deployment type
- Select Windows Installer (.msi file) and click Next.
- Browse and select the 7z2011-x64.msi file.
- Click Next.

Under Deployment type Wizard

Specify settings for this deployment type

Deployment type: Windows Installer (.msi file)

Installation file: 7z2011-x64.msi

Deployment type added.

6. CONFIGURE DEPLOYMENT TYPE

- User experience: Install for system
- Installation behavior: Install for system
- Reboot behavior: Suppress program notifications
- Click Next then Close.

User experience

Specify the user experience for this deployment type

User experience: Install for system

Installation behavior: Install for system

Reboot behavior: Suppress program notifications

Deployment type configured.

7. DISTRIBUTE CONTENT

- In the application, go to the Content tab.
- Click Distribute Content.
- Select Distribution Point(s).
- Click Next > Next > Close.

Distribute Content Wizard

Content Destination

Content will be distributed to the following distribution points, distribution point groups, and the distribution point group. Please verify internet connectivity.

Content destination: UNDISC\MEMCM\company.com

Content distributed to DP.

8. CREATE DEVICE COLLECTION

- Go to Assets and Compliance > Overview > Device Collections.
- Click Create Device Collection.
- Name: Win10-Clients
- Add members (or use query members) >
- Click OK.

Device Collection

Name: Win10-Clients

Linking Collection: All Systems

Member Count: 3

Device collection created.

9. DEPLOY APPLICATION

- Right-click 7-Zip application > Deploy.
- Select collection: Win10-Clients
- Action: Install
- Purpose: Required
- Click Next > Next > Close.

Deploy Software Wizard

General

Collection: Win10-Clients

Action: Install

Purpose: Required

Deployment created.

10. MONITOR DEPLOYMENT

- Go to Monitoring > Overview > Deployments
- Select the 7-Zip deployment.
- Review the overall compliance.

Deployments > Items

Item: 7-Zip

7-Zip Deployment to Win10-Clients

Compliance: 100% Compliant

Deployment success: 100% Compliant.

11. VERIFY ON CLIENT

- On Windows 10 client, open Command Prompt.
- Run 7z
- Verify 7-Zip is installed.

Command Prompt

```
C:\>7z
7-Zip 23.01 (64-bit)
Copyright (c) 1999-2023 Igor Pavlov

Usage: 7z [switches] [-wdir:dir] [-xfile_names...]

Comments:
  - Add files to archive
  - Extract files from archive
  - List contents of archive
  - ...
```

7-Zip installed on client.

12. REVIEW APPLICATION STATUS

- Go to Monitoring > Overview > Applications.
- Select 7-Zip.
- Review client installation status.

7-Zip Items

Item	Device Name	User	Status
7-Zip	WIN10-01	SYSTEM	Installed
7-Zip	WIN10-02	SYSTEM	Installed

Application installed on all clients.

12. SUMMARY

- Application created and content distributed
- Deployed to device collection
- Deployment successful
- 7-Zip installed on all clients
- Compliance 100%

VERIFICATION CHECKLIST

- Content distributed to DP
- Application deployment created
- Deployment shows 100% compliant
- 7-Zip installed on clients
- Application status shows Installed
- Logs show success (see below)

USEFUL SCCM LOGS

Log File Location	Purpose
Application.log (C:\Windows\CCM\Logs)	Application deployment and installation details
EAS.log (C:\Windows\CCM\Logs)	Client assignment and evaluation
ContentDownloaderManager.log (C:\Windows\CCM\Logs)	Content download from DP
LocationServices.log (C:\Windows\CCM\Logs)	Client location and boundary
Distmgr.log (C:\Program Files\Microsoft Configuration Manager\Logs)	Content distribution on DP
PsftHlp.log (C:\Program Files\Microsoft Configuration Manager\Logs)	Package transfer manager

COMMON ISSUES AND FIXES

Issue	Fix
Content not distributed	Check DP boundaries and Distribution Point Group
Deployment stuck in "In Progress"	Review Application.log and EAS.log on client
Installation failed	Check install requirements, requirements and disk space
Application not showing as installed	Verify collection method in Deployment Type
Clients not getting policy	Force policy on client: ccmexec /policy

KEY POINTS

- SCCM is used to deploy and manage software at scale.
- Applications are packaged, distributed and deployed to collections.
- Monitor deployments to ensure success.
- Logs are key to troubleshooting.
- Always verify on client and in the console.

Note: Lab environment uses a single Primary Site (MEM). Ensure clients have the SCCM client installed and are in the Win10-Clients collection.

LAB 15: SCCM SOFTWARE DEPLOYMENT (MSI APPLICATION)

Scenario: Deploy Google Chrome (MSI) to Windows 10 clients using SCCM and monitor the deployment.

1. LAB ENVIRONMENT

Site Code : NISM
SCCM Server : NISMCM (192.168.1.10)
Primary Site : NISM - NISMCM Primary Site
Console : Configuration Manager Console
Client OS : Windows 10 (21H2)
Domain : company.com
Tool : Microsoft SCCM (ConfigMgr)
Purpose : Deploy Google Chrome (MSI)

NETWORK DIAGRAM

2. DOWNLOAD GOOGLE CHROME MSI

- Go to <https://www.google.com/chrome/desktop/>
- Download Google Chrome MSI (64-bit).
- Save the file to a local folder.

Google Chrome MSI
File: GoogleChromeEnterprise.msi
Size: ~ 85 MB (approx.)

MSI downloaded.

3. CREATE APPLICATION

- In SCCM Console, go to: Software Library > Overview > Applications
- Click Create Application
- Select Manually specify the application information
- Click Next.

Application creation started.

4. GENERAL INFORMATION

- Name: Google Chrome
- Administrator comments: Google Chrome MSI
- Publisher: Google LLC
- Software version: 125.0.6422.76
- Click Next

Application details entered.

5. DEPLOYMENT TYPE

- Click Add > New deployment type
- Select Windows Installer (*.msi file) and click Next
- Browse and select GoogleChromeEnterprise.msi
- Click Next

Deployment type created.

6. USER EXPERIENCE

- Installation behavior: Install for systems
- Restart behavior: Suppress program notifications
- Installation program visibility: Normal
- Click Next

User experience configured.

7. REQUIREMENTS

- Click Add > New Requirement.
- Type: Operating System
- Operator: Equals
- Value: Windows 10 21H2
- Click OK and Next.

Requirements added.

8. DETECTION METHOD

- Add rule to detect Chrome installation.
- Rule type: MSI Product Code
- Value: {8A69D545-6664-483C-8F71-46909553994}
- Click Next.

Detection method configured.

9. CONTENT

- Content location: On this computer.
- Click Next.
- Click Close on completion.

Content configured.

10. SUMMARY

- Review the summary.
- Click Next.
- Click Close.

Application created successfully.

11. DEPLOY APPLICATION

- Right-click Google Chrome > Deploy.
- Select Collection: Win10-Clients
- Action: Install
- Purpose: Required
- Click Next > Next > Close.

Deployment created.

12. MONITOR DEPLOYMENT

- Go to Monitoring > Overview > Deployments
- Select the deployment.
- Check progress and status.

Deployment success: 100% Compliant.

13. VERIFY ON CLIENT

- On Windows 10 client, open Google Chrome.
- Go to <chrome://settings/help>
- Verify version is 125.0.6422.76 or higher.

Chrome installed and verified.

14. APPLICATION STATUS (CLIENT)

- On client, open Software Center.
- Go to Installation Status.
- Verify Google Chrome status.

Status: Installed

15. VERIFICATION CHECKLIST

- Application created in SCCM
- Deployment type created (MSI)
- Content distributed to DP
- Application deployed to collection
- Deployment success 100%
- Application installed on clients
- Application detected correctly
- Client status shows installed

16. USEFUL SCCM LOGS

Log File (Location)	Purpose
Applications.log (C:\Windows\CCM\Logs)	Application deployment and enforcement details
AppDiscovery.log (C:\Windows\CCM\Logs)	Application detection information
ContentTransferManager.log (C:\Windows\CCM\Logs)	Content download from DP
LocationServices.log (C:\Windows\CCM\Logs)	Client location and boundary
BrmgMgr.log (C:\Program Files\Microsoft Configuration Manager\Logs)	Content distribution via DP
PkgMgr.log (C:\Program Files\Microsoft Configuration Manager\Logs)	Package transfer manager

17. COMMON ISSUES AND FIXES

Issue	Fix
Content not distributed	Check BP, boundaries and Distribution Point Group
Deployment stuck in "In Progress"	Review AppDiscovery.log and CCM.log on client
Installation failed	Check install arguments, requirements and disk space
Application not detected	Verify detection rule and log AppDiscovery.log
Client not getting policy	Force policy on client <code>conexm /policy</code>

KEY POINTS

- SCCM is used to deploy MSI applications at scale.
- Applications are created, distributed and deployed to collections.
- Detection method ensures accurate compliance.
- Monitoring deployments ensures successful delivery.
- Logs help in troubleshooting and verifying deployments.

Note: Ensure clients have SCCM client installed, are in the Win10-Clients collection and can communicate with the management point.

LAB 16: SCCM OPERATING SYSTEM DEPLOYMENT (TASK SEQUENCE)

Scenario: Deploy Windows 10 22H2 to Windows 10 clients using SCCM Task Sequence and monitor the deployment.

1. LAB ENVIRONMENT

Site Code: MEM
 SCCM Server: MEMCM (192.168.1.10)
 Primary Site: MEM - MEMCM Primary Site
 Console: Configuration Manager Console
 Client OS: Windows 10 (21H2)
 Domain: company.com
 Tool: Microsoft SCCM (ConfigMgr)
 Task: Deploy Windows 10 22H2 using Task Sequence

NETWORK DIAGRAM

SCCM Server
MEMCM
192.168.1.10

Windows 10
Clients

SQL Server
(SCCM DB)

2. IMPORT OS IMAGE

- Go to Software Library > Overview > Operating Systems > Operating System Images.
- Click Import Operating System Image.
- Browse to the Windows 10 22H2 WIM file.
- Select WIM file and click Next.
- Enter image details and click Next.
- Complete the wizard.

Import Operating System Image Wizard

Image details

- Name: Windows 10 22H2
- Version: 10.0.19045
- Architecture: x64
- Language: en-US

Buttons: Previous, Next, Cancel

Operating System Image imported.

3. CREATE TASK SEQUENCE

- Go to Software Library > Overview > Operating Systems > Task Sequences.
- Click Create Task Sequence.
- Enter Task Sequence ID and Name.
- Select Boot Image.
- Choose Install Windows 10 22H2 image.
- Click Next.

Create Task Sequence Wizard

General

Task Sequence ID: W10-22H2-TS
 Task Sequence Name: Windows 10 22H2 TS

Buttons: Previous, Next, Cancel

Task Sequence created.

4. CONFIGURE TASK SEQUENCE STEPS

- In the Task Sequence Editor, add steps.
- Typical steps:
 - Apply Operating System Image
 - Apply Windows Settings
 - Apply Network Settings
 - Setup Windows and Configuration Manager
 - Enable BitLocker (Optional)
 - Restart Computer
- Click OK.

Buttons: Previous, Next, Cancel

Task Sequence steps configured.

5. DEPLOY TASK SEQUENCE

- Right-click the Task Sequence > Deploy.
- Select Collection: W10-Client.
- Click Next.
- Select purpose: Required.
- Click Next > Next > Close.

Deploy Software Wizard

General

Collection: W10-Client

Purpose: Required

Buttons: Previous, Next, Cancel

Task Sequence deployed.

6. DISTRIBUTE CONTENT

- Go to Software Library > Overview > Operating Systems > Task Sequences.
- Right-click the Task Sequence > Distribute Content.
- Select Distribution Points.
- Click Next > Next > Close.

Distribute Content Wizard

General

Content Distribution

Distribution points in distribution point group

Buttons: Previous, Next, Cancel

Content distributed.

7. BOOT CLIENT TO PXE

- Ensure the client is in W10-Client collection.
- Boot the client and press F12.
- Select PXE Boot.
- Client downloads policy and starts Task Sequence.

Press F12 for network service boot

Network boot from IP: 192.168.1.10
 CLIENT MAC ADDR: 08-00-56-08-3A-01 BIOS: 01
 CLIENT IP: 192.168.1.55 HIDE: 255.255.255.0
 GATEWAY IP: 192.168.1.1
 Download OS: 0000
 Press F12 for BIOS Menu
 Press F12 for network service boot
 Windows is loading files...

PXE boot initiated.

8. TASK SEQUENCE PROGRESS

- Task Sequence runs automatically.
- Steps are executed in order.
- Client reboots automatically when needed.
- Do NOT interrupt the process.

Installation Progress

MIN Company

Running action: Apply Operating System Image

Overall progress: 100%

Current step: Applying Windows 10 22H2

Elapsed time: 00:17:34 Remaining time: 00:10:45

Task Sequence in progress.

9. OS DEPLOYMENT COMPLETION

- Client restarts after installation.
- Windows setup and configuration complete.
- Client joins the domain and gets update.
- SCCM client installs automatically.

This might take several minutes.

OS deployment completed.

10. VERIFY CLIENT

- Log on to the client.
- Verify OS version.
- Verify SCCM client installed.
- Verify domain membership.

System

Edition: Windows 10 Pro
 Version: 22H2
 OS build: 19045.4331
 System type: 64-bit Operating System

Configuration Manager Properties

Client version: 5.00.9126.1900
 Client state: Active
 Site code: MEM

Client verified.

11. MONITOR DEPLOYMENT

- Go to Monitoring > Overview > Deployments.
- Filter by the Task Sequence.
- Check success, in progress or errors.

Deployments: 1 Item

Item	Software	Collection	Purpose	Completion %
Windows 10 22H2 TS	W10-Client	Required	100.0%	

Windows 10 22H2 TS to W10-Client

Summary: 2
 In Progress: 0
 Error: 0
 Unknown: 0

Total Assets: 2 (100.0% Success)

Deployment status: 100% Success.

12. CLEAN UP (OPTIONAL)

- Remove old boot images (if not used).
- Remove old OS images (if not used).
- Archive old Task Sequence (optional).
- Keep backups and logs.

Operating System Images

Item	Name	Version	Architecture
Windows 10 22H2	10.0.19045	x64	
Windows 10 21H2 (Old)	10.0.19044	x64	

Cleanup completed.

13. SUMMARY

- OS Image imported
- Task Sequence created and configured
- Task Sequence deployed to collection
- Content distributed to DP
- Clients boot via PXE
- Task Sequence executed successfully
- Windows 10 22H2 deployed
- Deployment 100% successful

SCCM Server
MEMCM
Deploy

W10-Client Collection

14. VERIFICATION CHECKLIST

- OS Image imported successfully
- Task Sequence created
- Content distributed to DP
- Task Sequence deployed to W10-Client
- Clients booted via PXE
- OS deployed successfully
- Client joined domain
- SCCM client installed and Active
- Deployment shows 100% Success

15. USEFUL SCCM LOGS

Log File (Location)	Purpose
smsts.log (C:\Windows\ConfigMgr\Logs)	Task Sequence execution details on client
adsetup.log (C:\Windows\ConfigMgr\Logs)	OS deployment steps and status
BDI.log (C:\Windows\ConfigMgr\Logs)	PXE boot and boot image logs
distribmgr.log (C:\Program Files\Microsoft Configuration Manager\Logs)	Content distribution to DP
smsexec.log (C:\Program Files\Microsoft Configuration Manager\Logs)	Deployment execution and status
cslog.log (C:\Windows\ConfigMgr\Logs)	Client assignment and evaluation

16. COMMON ISSUES AND FIXES

Issue	Fix
PXE boot fails	Check DHCP, PXE services, boot image and firewall
Content not available	Check DP status and content distribution
Task Sequence fails	Review smsts.log and adsetup.log
Client not showing in deployment	Check collection membership and client policy
OS not applying	Verify image, drivers and requirements

KEY POINTS

- Task Sequence automates OS deployment.
- Ensure boot image and OS image are correct.
- Distribute content to reachable DPs.
- Monitor deployment to ensure 100% success.
- Always review logs for troubleshooting.
- Keep OS and drivers up to date.

Note: Ensure clients have SCCM client installed, are in the W10-Client collection, and can communicate with Management Point and Distribution Points.

LAB 17: SCCM DRIVER DEPLOYMENT

Scenario: Import, package and deploy a device driver to Windows 10 clients using SCCM and monitor the deployment.

1. LAB ENVIRONMENT

Site Code : MEM
 SCCM Server : MEMCM (192.168.1.10)
 Primary Site : MEMCM - MEMCM Primary Site
 Console : Configuration Manager Console
 Client OS : Windows 10 (21H2)
 Domain : company.com
 Tool : Microsoft SCCM (ConfigMgr)
 Purpose : Deploy a Driver (Example: Realtek NIC Driver) to Win10 clients

NETWORK DIAGRAM

2. DOWNLOAD DRIVER

1. Download Realtek Ethernet Driver (64-bit).
 2. Save to local folder.

Realtek_NIC_Driver_10.61.zip
 Size: 25 MB (approx.)

Driver downloaded.

3. IMPORT DRIVER PACKAGE

1. Go to Software Library > Overview > Operating Systems > Driver Packages.
 2. Click Import Driver Package.
 3. Browse and select the driver folder.
 4. Click Next.

Import Driver Package Wizard

General | Data Source | Summary | Progress | Completion

Driver package source folder: D:\Drivers\Realtek_NIC
 Package source type: Driver package

Driver package imported.

4. DRIVER PACKAGE DETAILS

Driver Package Properties

Name: Realtek NIC Driver Package
 Driver Model: Real Drivers
 Architecture: x64
 Package ID: MEM0001A
 Created: 5/25/2024 11:05 AM
 Source: D:\Drivers\Realtek_NIC
 Status: Ready

Driver package ready.

5. CREATE DRIVER CATEGORY

1. Go to Software Library > Overview > Operating Systems > Driver Categories.
 2. Click Create Category.
 3. Name: Network Drivers.
 4. Click OK.

Create Category

Category name: Network Drivers
 Category description: Network adapters related drivers.

Driver category created.

6. IMPORT DRIVERS TO CATEGORY

1. Right-click Network Drivers > Import Driver.
 2. Select Realtek NIC Driver package.
 3. Select all models.
 4. Click OK.

Import Drivers

Available drivers: 1
☒ Realtek PCIe GBE Family Controller (x64)

Drivers added to category.

7. CREATE DEVICE COLLECTION

1. Go to Assets and Compliance > Overview > Device Collections.
 2. Click Create Device Collection.
 3. Name: Win10-Clients.
 4. Limiting Collection: All Systems.
 5. Click OK.

Device Collection

Name: Win10-Clients
 Limiting Collection: All Systems
 Member Count: 2

Device collection created.

8. CREATE DRIVER DEPLOYMENT

1. Right-click the Driver Category > Deploy.
 2. Select Collection: Win10-Clients.
 3. Action: Install.
 4. Purpose: Required.
 5. Click Next > Next > Close.

Deploy Drivers Wizard

Collection: Win10-Clients
 Action: Install
 Purpose: Required
 Driver category: Network Drivers

Driver deployment created.

9. DEPLOYMENT PROGRESS

1. Monitoring > Overview > Deployments

Deployments

Name	Software	Collection	Purpose	Compliance %
Realtek NIC Driver	Win10-Clients	Required	100.0%	

Realtek NIC Driver to Win10-Clients

Statistics

Status	Count	Percentage
Success	2	100%
In Progress	0	0%
Error	0	0%
Unknown	0	0%

Total Assets: 2 (100.00% Compliant)

Deployment successful.

10. VERIFY ON CLIENT

1. On a Windows 10 client, open Device Manager.
 2. Expand Network adapters.
 3. Verify Realtek adapter driver version.

Device Manager

Network adapters
☒ Realtek PCIe GBE Family Controller (Driver Version: 10.61.0514.2023)
☒ Intel(R) Wireless-AC 9560

Driver installed on client.

11. CHECK DRIVER COMPLIANCE

1. Go to Monitoring > Overview > Driver Status.
 2. Filter by Driver Package.
 3. Verify compliance is 100%.

Driver Status

Realtek NIC Driver Package

Compliant: 2
 Non-Compliant: 0
 Unknown: 0

Total Assets: 2 (100.00% Compliant)

Driver compliance 100%.

12. CLEAN UP (OPTIONAL)

1. Remove old driver packages.
 2. Remove unused driver categories.
 3. Keep backups and logs.

Driver Packages

Search:

Icon	Package Name	ID
☒	Realtek NIC Driver Package	MEM0001A

Cleanup completed.

13. SUMMARY

Win10-Clients Collection

- Driver package imported.
- Drivers added to category.
- Deployment created to Win10 clients.
- Driver installed on all clients.
- Compliance 100%.

14. VERIFICATION CHECKLIST

- Driver package imported successfully
- Drivers added to category
- Driver deployment created
- Deployment shows 100% compliant
- Driver installed on client
- Version verified in Device Manager
- Client communication successful
- Logs reviewed

15. USEFUL SCCM LOGS

Log File (Location)	Purpose
smsts.log (C:\Windows\CCM\Logs)	Driver package download and installation details
DriverInstall.log (C:\Windows\CCM\Logs)	Driver installation process on client
ContentTransferManager.log (C:\Windows\CCM\Logs)	Content download from DP
LocationServices.log (C:\Windows\CCM\Logs)	Location services and boundary
Distro.log (C:\ProgramData\Microsoft Configuration Manager\Logs)	Content distribution on DP

16. COMMON ISSUES AND FIXES

Issue	Fix
Driver not installing on client	Check driver compatibility and model match
Driver status shows Non-Compliant	Verify driver version and re-deploy if needed
Content not available on DP	Check DP, boundaries and content source
Client not getting policy	Force policy on client: ccmexec /policy
Driver category empty	Re-import drivers into category

17. KEY POINTS

- Driver packages make driver deployment easy and reliable.
- Use categories to organize drivers.
- Deploy to specific collections.
- Monitor compliance to ensure success.
- Always verify on client.
- Keep driver packages and logs safe.

Note: Ensure clients have SCCM client installed, are in the correct collection, and can communicate with Management Point and Distribution Points.

LAB 18: SCCM OS DEPLOYMENT USING TASK SEQUENCE

Scenario: Deploy Windows 10 22H2 to Windows 10 clients using SCCM Task Sequence and monitor the deployment.

1. LAB ENVIRONMENT

Site Code : MEM
 SCCM Server : MEMCM (192.168.1.10)
 Primary Site : MEMCM Primary Site
 Console : Configuration Manager Console
 Client OS : Windows 10 (21H2)
 Domain : company.com
 Tool : Microsoft SCCM (ConfigMgr)
 Purpose : Deploy Windows 10 22H2 using Task Sequence

NETWORK DIAGRAM

SCCM Server MEMCM 192.168.1.10

Windows 10 Clients

SQL Server (SCCM DB)

2. IMPORT OS IMAGE

- Go to Software Library > Overview > Operating Systems > Operating System Images.
- Click Import Operating System Image.
- Browse to the Windows 10 22H2 WIM file.
- Select WIM file and click Next.
- Enter image details and click Next.
- Complete the wizard.

Import Operating System Image Wizard

Summary

Image details

- Name: Windows 10 22H2
- Version: 10.0.19045
- Architecture: x64
- Language: en-US

Next > Cancel

OS Image imported.

3. CREATE TASK SEQUENCE

- Go to Software Library > Overview > Operating Systems > Task Sequences.
- Click Create Task Sequence.
- Enter Task Sequence ID and Name.
- Select Boot image.
- Choose Install Windows 10 22H2 image.
- Click Next.

Create Task Sequence Wizard

General

Task Sequence ID: W10-22H2-TS

Task Sequence Name: Windows 10 22H2 TS

Boot image: BootImage(x64)

OS Image: Windows 10 22H2 (x64)

Next > Cancel

Task Sequence created.

4. CONFIGURE TASK SEQUENCE STEPS

- In the Task Sequence Editor, add steps.
- Typical steps:
 - Apply Operating System Image
 - Apply Windows Settings
 - Apply Network Settings
 - Setup Windows and Configuration Manager
 - Enable BitLocker (Optional)
 - Restart Computer
- Click OK.

Task Sequence: Windows 10 22H2 TS

Installation

- Windows 10 22H2 TS
- State Capture
- Postinstall
- Install Operating System
- Apply Operating System Image
- Apply Windows Settings
- Setup Windows and Configuration Manager
- Enable BitLocker (Optional)
- Restart Computer
- Monitor

Task Sequence steps configured.

5. DEPLOY TASK SEQUENCE

- Right-click the Task Sequence > Deploy.
- Select Collection: Win10-Clients.
- Click Next.
- Select purpose: Required.
- Click Next > Next > Close.

Deploy Task Sequence Wizard

General

Task Sequence: Windows 10 22H2 TS

Collection: Win10-Clients

Purpose: Required

Next > Cancel

Task Sequence deployed.

6. DISTRIBUTE CONTENT

- Go to Software Library > Overview > Operating Systems > Task Sequences.
- Right-click the Task Sequence > Distribute Content.
- Select Distribution Points.
- Click Next > Next > Close.

Distribute Content Wizard

General

Content Distribution

Distribution Points or Distribution Point Groups

Summaries

- MEMCM (192.168.1.10)
- MEMCM-Client

Next > Cancel

Content distributed.

7. BOOT CLIENT TO PXE

- Ensure the client is in Win10-Clients collection.
- Boot the client and press F12.
- Select PXE Boot.
- Client downloads policy and starts Task Sequence.

Press F12 for network service boot

Network boot from Intel i386

CLIENT PC: 192.168.1.15 - BIOS: 0.00.01

CLIENT IP: 192.168.1.15 - MAC: 25.25.25.9

GATEWAY IP: 192.168.1.1

Downloaded BIOSMP...

Press F8 for BIOS Menu

Press F12 for network service boot

Windows is loading files...

PXE boot initiated.

8. TASK SEQUENCE PROGRESS

- Task Sequence runs automatically.
- Steps are executed in order.
- Client reboots automatically when needed.
- Do NOT interrupt the process.

Installation Progress

MEM Company

Running action: Apply Operating System Image

Overall progress: 100%

Current step: Apply Windows 10 22H2 Image

Elapsed time: 00:15:30 Remaining time: 00:05:10

Task Sequence in progress.

9. OS DEPLOYMENT COMPLETION

- Client restarts after installation.
- Windows setup and configuration complete.
- Client joins the domain and gets policy.
- SCCM client installs automatically.

This might take several minutes

Don't turn off your PC

OS deployment completed.

10. VERIFY CLIENT

- Log on to the client.
- Verify Windows 10 version.
- Verify SCCM client installed.
- Verify domain membership.

System

Edition: Windows 10 Pro

Version: 22H2

OS build: 19045.22H2

System type: 64-bit Operating System

Configuration Manager Properties

Client version: 5.00.9135.1000

Client state: Active

Site code: MEM

Client verified.

11. MONITOR DEPLOYMENT

- Go to Monitoring > Overview > Deployments.
- Filter by the Task Sequence.
- Check success, in progress or errors.

Deployments: 1 items

Search

Task Sequence: Windows 10 22H2 TS

Collection: Win10-Clients

Purpose: Required

Compliance: 100.0%

Statistics

Windows 10 22H2 TS to Win10-Clients

Success: 2

In Progress: 0

Error: 0

Unknown: 0

Total Assets: 2 (100.00% Success)

Deployment status: 100% Success.

12. CLEAN UP (OPTIONAL)

- Remove old boot images (if not used).
- Remove old OS images (if not used).
- Archive old Task Sequence (optional).
- Keep backups and logs.

Operating System Images

Search

Task Sequence: Windows 10 22H2 TS

Collection: Win10-Clients

Purpose: Required

Compliance: 100.0%

Statistics

Windows 10 22H2 TS to Win10-Clients

Success: 2

In Progress: 0

Error: 0

Unknown: 0

Total Assets: 2 (100.00% Success)

Cleanup completed.

13. SUMMARY

SCCM Server MEMCM 192.168.1.10

Win10 Clients Collection

- OS Image imported
- Task Sequence created
- Task Sequence configured and deployed
- Content distributed to DPs
- Clients boot to PXE
- Task Sequence executed successfully
- Windows 10 22H2 deployed
- Deployment 100% successful

14. VERIFICATION CHECKLIST

- OS Image imported successfully
- Task Sequence created
- Content distributed to DP
- Task Sequence deployed to Win10-Clients
- Clients booted via PXE
- OS deployed successfully
- Client joined domain
- SCCM client installed and Active
- Deployment shows 100% Success

15. USEFUL SCCM LOGS

Log file (Location) Purpose

smsts.log (C:\Windows\CCM\Log) Task Sequence execution details on client

extsmsts.log (C:\Windows\CCM\Log) OS deployment steps and status

smsts.log (C:\Windows\CCM\Log) PXE boot and BIOS

extsmsts.log (C:\Windows\CCM\Log) Content distribution to DP

extsmsts.log (C:\Program Files\Microsoft Configuration Manager\Log) Deployment execution and status

extsmsts.log (C:\Program Files\Microsoft Configuration Manager\Log) Client deployment and validation

16. COMMON ISSUES AND FIXES

Issue	Fix
PXE boot fails	Check DHCP, PXE service, boot image and firewall
Content not available	Check DP status and content distribution
Task Sequence fails	Review smsts.log and extsmsts.log
Client not showing in deployment	Check collection membership and client policy
OS not applying	Verify image, drivers and requirements

KEY POINTS

- Task Sequence automates OS deployment.
- Ensure boot image and OS image are correct.
- Distribute content to reachable DPs.
- Monitor deployment to ensure 100% success.
- Always review logs for troubleshooting.
- Keep OS and drivers up to date.

Note: Ensure clients have SCCM client installed, are in the Win10-Clients collection, and can communicate with Management Point and Distribution Points.

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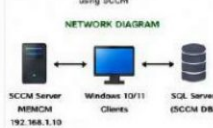
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LAB 19: SCCM DRIVER PACKAGE CREATION AND DEPLOYMENT

Scenario: Create a Driver Package for a Dell Latitude Laptop and deploy to Windows 10/11 clients using SCCM. Monitor and verify the deployment.

1. LAB ENVIRONMENT

Site Code: MEM
 SCCM Server: MEMCM (192.168.1.10)
 Primary Site: MEM - MEMCM (Primary Site)
 Console: Configuration Manager Console
 Client OS: Windows 10/11 (2H2/G2H2)
 Domain: company.com
 Tool: Microsoft SCCM (ConfigMgr)
 Purpose: Create and deploy Driver Package using SCCM



2. DOWNLOAD DRIVERS

- Go to Dell Support website.
- Enter your laptop model or Service Tag.
- Select Windows 10/11 (64-bit).
- Download all required driver categories.
- Extract the downloaded files.

Dell Support

Enter a URL product or Service Tag:

Drivers & Downloads

Operating System: Windows 10 64-bit

☒ Chipset Driver
☒ BIOS Management Engine Interface
☒ SerialIO Driver
☒ Display Driver
☒ Network Driver
☒ Audio Driver
☒ Others

Drivers downloaded and extracted.

3. CREATE DRIVER PACKAGE

- In SCCM Console, go to Software Library > Overview > Operating Systems > Driver Packages.
- Right-click Driver Packages > Create Driver Package.
- Enter Name, Description and select Driver Source Folder.
- Click Next > Summary > Next > Close.

Create Driver Package Wizard

General

Driver Source: Dell_Latitude_Drivers

Summary: Drivers for Dell Latitude Laptops

Progress: Driver source folder: \\MEMCM\Drivers\Dell_Latitude

Completion: \\MEMCM\Drivers\Dell_Latitude

Driver Package created successfully.

4. IMPORT DRIVER PACKAGE CONTENT

- The driver files are imported from the source folder.
- SCCM processes and categorizes the drivers.
- Wait for content to be distributed to DPs.

Driver Package Content

Total Drivers: 24

Categories:

- Chipset (2)
- System (2)
- Display (2)
- Network (2)
- Mouse, Keyboard & Input (2)
- Storage (2)
- Ports (2)
- Audio (2)
- Others (16)

Status: Content distribution in progress.

Driver content imported and distributing.

5. CREATE DEPLOYMENT

- Right-click the Driver Package > Deploy.
- Select Collection: Win10-Clients.
- Select purpose: Required.
- Choose deployment settings.
- Click Next > Next > Close.

Deploy Driver Package Wizard

General

Content: Dell_Latitude_Drivers

Deployment Design: Collection: Win10-Clients

Sharing: User Instance: Purpose: Required

Alerts: ☐ Use default distribution point group

Deployment created.

6. DRIVER DISTRIBUTION

- Driver content is distributed to Distribution Points.
- Monitor content status in SCCM.
- Ensure DP is reachable by clients.

Content Status to DP

Distribution Point: MEMCM (192.168.1.10)

Content	Status	% Complete
Driver Package	In Progress	75%

Driver content distribution in progress.

7. CLIENT POLICY & CONTENT

- Clients communicate with Management Point.
- Policy is downloaded and processed.
- Driver package is available for installation.
- Content is downloaded from DP.

Client Actions

Policy retrieval: Success

Driver assignment: Dell_Latitude_Drivers

Content download: In Progress

Location: MEMCM (192.168.1.10)

Point: 1

Download progress: 48%

Client is downloading driver content.

8. DRIVER INSTALLATION

- Driver package installs automatically.
- Installation runs in background.
- System may restart if required.
- Check compliance status.

Installation Status

Package: Dell_Latitude_Drivers

Status: Installing

Overall Progress: 80%

Last Update: 5/20/2024 10:30 AM

Estimated Time Remaining: 00:02:15

Drivers are installing on client.

9. VERIFY DRIVER INSTALLATION

- Open Device Manager.
- Verify drivers are installed and working.
- No unknown devices.

Device Manager (Sample)

- Computer
- Disk drives
- Display adapters
- Intel(R) UHD Graphics
- Network adapters
- Intel(R) Ethernet Controller
- Sound, video and game controllers
- Realtek audio
- System devices
- Universal Serial Bus controllers

Drivers installed successfully.

10. MONITOR DEPLOYMENT

- Go to Monitoring > Overview > Deployments.
- Filter by the deployment.
- Check success, in progress and errors.

Deployments: 1 items

Item	Software	Collection	Purpose	Compliance %
Dell_Latitude_Drivers	Win10-Clients	Required	100.0%	

Deployment Statistics

Success: 10
 In Progress: 0
 Error: 0
 Unknown: 0

Total Assets: 10 (100.0% Success)

Deployment status: 100% Success.

11. REPORTS (OPTIONAL)

- Go to Monitoring > Reports.
- Run: Drivers - Driver Compliance.
- View compliance summary.

Driver Compliance (Example)

Collection: Win10-Clients

Total Devices: 10

Compliant: 10 (100%)

Non-Compliant: 0 (0%)

Unknown: 0 (0%)

All clients are compliant.

12. CLEAN UP (OPTIONAL)

- Remove old or unused driver packages.
- Remove old content from DPs.
- Archive driver source if required.
- Keep backups and logs.

Driver Packages

Search:

Package Name	Created
Dell_Latitude_Drivers	Created: 5/20/2024
Old_Latitude_Drivers	Created: 3/15/2024
Test_Drivers	Created: 1/15/2024

Cleanup completed.

13. SUMMARY



- Driver Package created
- Content distributed to DPs
- Deployment created to collection
- Clients downloaded content
- Drivers installed successfully
- Deployment 100% successful

14. VERIFICATION CHECKLIST

- Driver package created successfully
- Driver content distributed to DPs
- Deployment created to collection
- Clients received policy
- Content downloaded by clients
- Drivers installed without errors
- No unknown devices in Device Manager
- Deployment success: 100%
- Clients are compliant

15. USEFUL SCCM LOGS

Log File (Location)	Purpose
smstslog (C:\Windows\CCM\Log)	Task sequence execution (if used for driver install)
dpdownloadlog (C:\Windows\CCM\Log)	Content download by client
dpngetlog (C:\Program Files\Microsoft Configuration Manager\logs)	Content distribution to DPs
pkgclntlog (C:\Program Files\Microsoft Configuration Manager\logs)	Package transfer manager
appenforce.log (C:\Windows\CCM\Log)	Driver installation and enforcement details
CAS.log (C:\Windows\CCM\Log)	Client assignment and evaluation

16. COMMON ISSUES AND FIXES

Issue	Fix
Content not distributed	Check DP boundaries and distribution point group
Clients not downloading content	Check DP connectivity, peer cache and firewall
Drivers not installing	Check driver compatibility and remove AppEnforce.log
Unknown device in Device Manager	Update drivers and re-run deployment
Deployment stuck	Review smsts.log and policy assignment

KEY POINTS

- Use vendor website to download correct drivers.
- Create a driver package and import all drivers.
- Distribute content to DPs before deployment.
- Deploy to the correct collection.
- Monitor deployment and verify compliance.
- Keep driver packages updated regularly.
- Driver automation improves client reliability.

Note: Ensure clients have SCCM client installed and can communicate with MP and DPs.

LAB 20: SCCM PATCH MANAGEMENT DEPLOYMENT

Scenario: Configure and deploy Software Updates to Windows 10/11 clients using SCCM and monitor compliance.

1. LAB ENVIRONMENT

Site Code: MINM
 SCCM Server: MINMCM (192.168.1.10)
 Primary Site: MINM - MINMCM Primary Site
 Console: Configuration Manager Console
 Client OS: Windows 10/11 (21H2)
 Domain: corpnet.com
 Tool: Microsoft SCCM (ConfigMgr)
 Purpose: Configure and deploy Software Updates

NETWORK DIAGRAM

2. CONFIGURE SOFTWARE UPDATE POINT

- Go to Administration > Overview > Site Configuration > Servers and Site System Roles.
- Select server and click Add Site System Role.
- Select 'Software Update Point' and click Next.
- Choose software update point role and click Next.
- Select synchronization and download settings.
- Click Next > Next > Close.

Software Update Point
 Specify the synchronization schedule.

Synchronization schedule: Daily
 Synchronization time: 03:00 AM

Download settings:
☒ Download software updates
☒ Download associated content

< Previous Next > Cancel

3. SYNC SOFTWARE UPDATES

- Go to Software Library > Overview > Software Updates > All Software Updates.
- Click 'Synchronize Software Updates'.
- Select 'Full Synchronization' and click OK.
- Monitor sync progress in the console.

Synchronization Progress
 Status: In Progress
 Progress: 72%

Statistics:
 Total updates: 3600
 Processed: 2592
 New updates: 528
 Expired/Declined: 0
 Errors: 0

Last Sync Time: 5/29/2024 03:00 AM

< Previous Next > Cancel

4. CREATE SOFTWARE UPDATE GROUP

- Go to Software Library > Overview > Software Updates > Software Update Groups.
- Click 'Create Software Update Group'.
- Enter Name and Description.
- Choose criteria to include updates.
- Click OK.

Create Software Update Group
 Name: Win10-11 Critical Updates
 Description: Critical and security updates for Windows 10/11

Update group criteria
 Classifiers: Critical updates, security updates
 Products: Microsoft Windows 10

OK Cancel

5. CREATE DEPLOYMENT PACKAGE

- Go to Software Library > Overview > Packages.
- Click 'Create Package'.
- Enter Name, Description, and Source folder.
- Click Next > Next > Close.

New Package
 Name: Win10-11 Updates Package
 Description: Package for Win10-11 updates
 Source folder: \\MEMCM\Source\Updates

< Previous Next > Close

6. DEPLOY SOFTWARE UPDATES

- Go to Software Library > Overview > Software Update Groups.
- Right-click 'Win10-11 Critical Updates' > Deploy Software Updates.
- Select Collection: Win10-11 Clients.
- Select Deployment Package.
- Choose schedule and user experience.
- Click Next > Next > Close.

Deployment Settings
 Collection: Win10-11 Clients
 Package: Win10-11 Updates Package
 Schedule: As soon as possible
 User experience: Install for system

< Previous Next > Close

7. CLIENT POLICY RETRIEVAL

- Client communicates with Management Point.
- Policy is downloaded and processed.
- Client evaluates assigned updates.
- Updates are scheduled for installation.

Client Actions
 Policy retrieval: Success
 Assigned updates: 12
 Evaluation time: 5/29/2024 03:15 AM
 Compliance state: Non-Compliant
 Next scan time: 5/29/2024 04:15 AM

Client policy received.

8. SOFTWARE UPDATE DOWNLOAD

- Content downloads from SUP to client.
- Content is stored in local cache.
- Download progress is monitored.

Download Status (Client)
 Total updates: 12
 Downloaded: 8
 Downloading: 2
 Pending: 2
 Progress: 66%

Updates are downloading.

9. INSTALL SOFTWARE UPDATES

- Updates are installed based on schedule.
- Client may require restart for some updates.
- Installation status is reported to SCCM.

Installation Status
 Updates installed: 8
 Updates pending: 2
 Updates failed: 0
 Reboot required: Yes
 Last installation time: 5/29/2024 04:30 AM

Updates installed successfully.

10. MONITOR DEPLOYMENT

- Go to Monitoring > Overview > Software Update Deployments.
- Select the deployment.
- Review overall compliance details.

Deployment Summary
 Collection: Win10-11 Clients
 Total Resources: 10
 Compliant: 8 (80%)
 In Progress: 1 (10%)
 Non-Compliant: 1 (10%)
 Unknown: 0 (0%)

Deployment status: 80% Compliant.

11. REPORTS (OPTIONAL)

- Go to Monitoring > Reports > Software Updates.
- Run compliance and installation reports.
- Review and export if needed.

Compliance Report (Sample)
 Collection: Win10-11 Clients
 Report: Software Update Compliance
 Generated: 5/29/2024 04:45 AM

Compliance State	Resources	%
Compliant	8	80%
Non-Compliant	1	10%
In Progress	1	10%
Unknown	0	0%

Reports generated.

12. CLEAN UP (OPTIONAL)

- Remove old software update deployments.
- Remove expired software updates.
- Clean up unused packages if required.
- Keep backups and logs.

Cleanup Actions
☒ Old deployments removed
☒ Expired updates declined
☒ Unused packages cleaned
☒ Logs archived

Cleanup completed.

13. SUMMARY

- SUP configured and synchronized
- Software Update Group created
- Deployment Package created
- Updates deployed to collection
- Clients downloaded and installed updates
- Compliance monitored and verified
- Reports generated
- Deployment 80% compliant

14. VERIFICATION CHECKLIST

- SUP role installed and synced
- Software Updates synchronized
- Software Update Group created
- Deployment Package created
- Clients received policy
- Updates downloaded and installed
- Deployment status monitored
- Reports reviewed
- Compliance at expected level

15. USUFL SCCM LOGS

Log File Location	Purpose
\\Win10-11\log (C:\Windows\CCM\Log)	Software update evaluation and detection
\\Win10-11\log (C:\Windows\CCM\Log)	Deployment of software updates
\\Win10-11\log (C:\Windows\CCM\Log)	Content download from SUP
\\Win10-11\log (C:\Program Files\Microsoft Configuration Manager\Logs)	Deployment execution on client
\\Win10-11\log (C:\Windows\CCM\Log)	Network management
\\Win10-11\log (C:\Windows\CCM\Log)	Scan cycles and compliance

16. COMMON ISSUES AND FIXES

Issue	Fix
Clients not receiving updates	Check client policy, MP connectivity and deployment targeting
Download failing	Check SUP BP connectivity and content availability
Updates not installing	Check Eventlog, client health and permissions
Reboot not happening	Check maintenance window and reboot settings
Compliance not reporting	Force policy and compare logs

KEY POINTS

- Configure SUP for downloading updates.
- Create update groups with proper criteria.
- Deploy updates to the correct collection.
- Monitor deployment and compliance regularly.
- Review reports and act on non-compliant systems.
- Keep systems up-to-date and secure.

NOTE: Ensure clients have SCCM client installed, can communicate with Management Point and Software Update Point, and have internet access for update synchronization.

LAB 21: SCCM BOUNDARY GROUP CONFIGURATION AND CLIENT ASSIGNMENT

Scenario: Configure Boundary Groups and Boundaries and assign SCCM clients to the appropriate boundary groups for content and policy location.

1. LAB ENVIRONMENT

Site Code : MINM
 SCCM Server : MINCM (192.168.1.10)
 Primary Site : MINM - MINCM Primary Site
 Console : Configuration Manager Console
 Client OS : Windows 10/11 (21H2/22H2)
 Domain : company.com
 Tool : Microsoft SCCM (ConfigMgr)
 Purpose : Configure Boundary Groups assign clients for location services

NETWORK DIAGRAM

2. CREATE BOUNDARY GROUPS

- Go to Administration > Overview > Site Configuration > Boundaries > Site Configuration > Boundaries > Create Boundary Group
- Right-click Boundary Groups > Create Boundary Group
- Create the following Boundary Groups

Boundary Group Name	Description
Headquarters-BG	Headquarters Office
BranchOffice-BG	Branch Office
RemoteOffice-BG	Remote Office

4. Click OK.

Create Boundary Group

Name: BranchOffice-BG
 Description: Branch Office

OK Cancel

Boundary Groups created.

3. CREATE BOUNDARIES

- Right-click the required Boundary Group > Create Boundary
- Select Boundary type (IP Address Range or IP Subnet)
- Add boundaries for each group as shown.

Boundary Group	Type	Start IP	End IP / Subnet
Headquarters-BG	IP Subnet	192.168.1.0	255.255.255.0
BranchOffice-BG	IP Subnet	192.168.2.0	255.255.255.0
RemoteOffice-BG	IP Subnet	192.168.3.0	255.255.255.0

4. Click OK.

Create Boundary

Type: IP Subnet
 Start IP: 192.168.2.0
 Subnet Mask: 255.255.255.0

OK Cancel

Boundaries created successfully.

4. CONFIGURE BOUNDARY GROUP RELATIONSHIP

- Go to Boundary Groups node
- Right-click each Boundary Group > Relationships
- Ensure the Boundary Groups are related in Hierarchy.

Boundary Group Relationship

4. Click OK.

Boundary Group relationship configured.

5. ASSIGN SITE SYSTEM ROLES TO BOUNDARY GROUPS

- Go to Administration > Overview > Site Configuration > Site
- Select Site > Configure Site Components > Software Distribution
- Add/Verify Distribution Points in each Boundary Group.

Boundary Group	Distribution Point
Headquarters-BG	MINCM (192.168.1.10)
BranchOffice-BG	DP-Branch (192.168.2.20)
RemoteOffice-BG	DP-Remote (192.168.3.20)

4. Click OK.

Distribution Point

Select boundary group: BranchOffice-BG
 Distribution point: DP-Branch (192.168.2.20)

OK Cancel

Distribution Points assigned.

6. ENABLE CLIENT SETTINGS

- Go to Administration > Overview > Client Settings
- Select or create a Client Setting
- Enable the following:
 - Enable client fallback
 - Site assignment by Boundary Groups
 - Allow clients to use a fallback site
- Click OK.

Client Setting

Enable client fallback ☒
 Site assignment: By Boundary Groups
 Allow clients to use a fallback site ☒

OK Cancel

Client settings enabled.

7. ASSIGN CLIENTS TO BOUNDARY GROUPS

- Go to Assets and Compliance > Devices
- Select the client device
- Right-click > Client Settings > Assign to Site
- Select Boundary Group.

Assign to Site

Assign the selected objects to the following site:

Boundary Group: BranchOffice-BG

OK Cancel

Clients assigned to Boundary Groups.

8. CLIENT LOCATION VERIFICATION

- On the client, open Command Prompt
- Run: ipconfig
- Verify IP address falls into correct Boundary.
- Run: ccmexec /location
- Verify the assigned site.

```
C:\> ipconfig
IPv4 Address . . . . . : 192.168.2.15
C:\> ccmexec /location
NDCLCompany.com (192.168.1.10)
Assigned site code: MIN
Assigned boundary group: BranchOffice-BG
IP Address Range: 192.168.2.0 - 192.168.2.255
```

Client location verified.

9. CONTENT LOCATION VERIFICATION

- On the client, open Command Prompt
- Run: ccmexec /download
- Verify content source is from correct DP

```
C:\> ccmexec /download
Content download location information:
Content ID : ABC12345
Content source : DP-Branch.Company.com (192.168.2.20)
Content size : 100 MB
Content path : C:\Windows\ccmcache\1
```

Content source verified.

10. MONITOR BOUNDARY GROUP STATUS

- Go to Monitoring > Overview > Boundary Status
- Verify boundaries and boundary group status
- All should be Up.

Boundary Group	Boundary	Type	Status
Headquarters-BG	192.168.1.0/24	IP Subnet	Up
BranchOffice-BG	192.168.2.0/24	IP Subnet	Up
RemoteOffice-BG	192.168.3.0/24	IP Subnet	Up

Boundary status is healthy.

11. REPORTS (OPTIONAL)

- Go to Monitoring > Reports > Site > Client Information
- Run the following reports:
 - Clients by Boundary Group
 - Clients with No Boundary Group
 - Site System Status

Clients by Boundary Group

Reports generated.

12. CLEAN UP (OPTIONAL)

- Remove test clients or unassign if needed.
- Remove test boundaries if no longer required.
- Remove test Distribution Points (optional).
- Archive reports and logs.

Cleanup Summary

- Test clients unassigned
- Test boundaries removed
- Test DPs removed (if any)
- Reports and logs archived

Cleanup completed.

13. SUMMARY

- Boundary Groups created
- Boundaries created and related
- Distribution Points assigned
- Clients assigned to correct Boundary Groups
- Content source verified
- Client location verified
- Distribution and policy working
- Deployment 100% successful

14. VERIFICATION CHECKLIST

- Boundary Groups exist and related
- Boundaries are correct and Up
- DPs assigned to each Boundary Group
- Clients assigned to correct Boundary Group
- Clients can reach MP
- Content downloaded from correct DP
- Policy received successfully
- All monitoring status is healthy

15. USER'S SCCM LOGS

Log File Location	Purpose	Issue	Fix
LocationServices.log (C:\Windows\CCM\Log)	Boundary discovery and client location assignment	Client not assigned to correct Boundary Group	Verify IP, boundaries and Group relationship
Multicast.log (C:\Windows\CCM\Log)	MP location and selection	Client shows Unknown Site	Ensure MP reachable and boundaries correct
BoundaryTransferService.log (C:\Windows\CCM\Log)	Content download and transfer	Content not downloading	Verify DP assignment and content status
Enrollment.log (C:\ProgramData\Microsoft\ContentManager\Logs)	Content distribution status	Boundary shows Down	Check site system, firewall and MP connectivity
ClientManagerStartup.log (C:\Windows\CCM\Log)	Client start and site assignment	Policy not received	Force policy and check logs

16. COMMON ISSUES AND FIXES

Issue: Client not assigned to correct Boundary Group

Fix: Verify IP, boundaries and Group relationship

Issue: Client shows Unknown Site

Fix: Ensure MP reachable and boundaries correct

Issue: Content not downloading

Fix: Verify DP assignment and content status

Issue: Boundary shows Down

Fix: Check site system, firewall and MP connectivity

Issue: Policy not received

Fix: Force policy and check logs

KEY POINTS

- Boundary Groups control site assignment.
- Clients get policy and content from nearest DP.
- Hierarchical relationship ensures fallback.
- Always verify location and content source.
- Monitoring and logs help in troubleshooting.
- Keep boundaries and DPs updated.

NOTE: Ensure clients have SCCM client installed, correct IP configuration and can communicate with Management Point and Distribution Points.

WINDOWS SERVER ADMINISTRATION – END TO END LAB

Scenario: Install, configure and manage a Windows Server 2022 system and provide core services for client computers.

1. LAB ENVIRONMENT

Server OS
 - Windows Server 2022 Standard
Hostname
 - SRV-DC01
IP Address
 - 192.168.1.10/24
Domain
 - company.com
DNS Server
 - 192.168.1.10 (local)
Gateway
 - 192.168.1.1
Client OS
 - Windows 10/11
Client IP Range
 - 192.168.1.50 - 200

NETWORK DIAGRAM

2. INSTALL WINDOWS SERVER

- Boot from Windows Server 2022 ISO.
- Select language, time and keyboard.
- Click Install Now.
- Select Windows Server 2022 Standard.
- Accept license terms and click Next.
- Choose Custom: Install Windows only.
- Select disk, click Next and install.
- After installation, set Administrator password.
- Complete initial configuration.

Windows Server installed.

3. INITIAL CONFIGURATION

- Reboot the server to SRV-DC01.
- Set static IP: 192.168.1.10/24, GW: 192.168.1.1.
- Set DNS to 127.0.0.1 or 192.168.1.10.
- Update the server.
- Set time zone and time.
- Enable Remote Desktop.
- Configure Windows Firewall (Domain Profile).

Initial configuration completed.

4. PROMOTE TO DOMAIN CONTROLLER (AD DS)

- Open Server Manager > Add Roles and Features.
- Select Role-based installation.
- Select this server: SRV-DC01.
- Select Active Directory Domain Services.
- Add required features and click Next.
- Install and close the wizard.
- Click the flag > Promote this server to a domain controller.
- Select Add a new forest.
- Enter Root domain name: company.com.
- Enter DSRM password and click Next.
- Review and click Install.
- Server will restart.

Server promoted to Domain Controller.

5. CONFIGURE DNS

- DNS role is installed automatically with AD DS.
- Verify forward lookup zone: company.com.
- Verify reverse lookup zone: 1.168.192.in-addr.arpa.
- Ensure DNS is listening on the server IP.

DNS configured.

6. CREATE OUs AND USERS

- Open Active Directory Users and Computers.
- Create the following OUs in company.com:
 - Departments
 - Users
 - Groups
- Create Users:
 - john (OU: Users)
 - mary (OU: Users)
- Set passwords and uncheck "User must change password at next login".
- Create Groups:
 - IT Users (OU: Users)
 - HR Users (OU: Users)
- Add john to IT Users and mary to HR Users.

OU's, Users and Groups created.

7. CONFIGURE DHCP SERVER

- Add DHCP Server role.
- Authorize DHCP in AD.
- Create IP4 Scope:
 - Scope Name: LAN-SCOPE
 - IP Range: 192.168.1.50 - 192.168.1.200
 - Subnet Mask: 255.255.255.0
 - Default Gateway: 192.168.1.1
 - DNS Server: 192.168.1.10
- Activate the scope.

DHCP configured and activated.

8. CONFIGURE FILE SERVICES AND SHARES

- Add File and Storage Services role.
- Create a new volume (if required) or use existing.
- Create folder: D:\Share\CompanyData
- Right-click > Properties > Sharing > Advanced Sharing.
- Share name: CompanyData
- Set NTFS Permissions:
 - IT Users - Modify
 - HR Users - Read
 - Administrators - Full Control
- Verify share permissions.

File share created and permissions set.

9. GROUP POLICY BASICS

- Open Group Policy Management.
- Right-click company.com > Create a GPO.
- Name: Default Domain Policy - Custom.
- Edit the GPO and configure:
 - Password Policy
 - Desktop Background
 - Disable Control Panel
- Link the GPO to the domain.
- Run gpupdate /force on client.

GPO created and applied.

10. JOIN CLIENTS TO DOMAIN

- On client, set DNS to 192.168.1.10.
- Right-click This PC > Properties > Change settings.
- Click Change > Domain > company.com.
- Enter domain admin credentials.
- Restart the client.
- Login with domain user (e.g. john).
- Verify access to \\SRV-DC01\CompanyData.

Client joined to domain.

11. MONITOR AND MAINTAIN

- Check Event Viewer for errors.
- Monitor Services and Performance.
- Backup AD DS using Windows Server Backup.
- Verify DHCP issues and DNS records.
- Use Server Manager Dashboard for health.

Server is healthy and running.

12. CLEAN UP (LAB END)

- Remove test users and groups.
- Remove test OUs if not needed.
- Remove file shares.
- Delete DHCP scope (if required).
- Remove client from domain.
- Take final backup and logs.

Cleanup completed.

13. SUMMARY

- Windows Server installed and configured
- Active Directory Domain Services installed
- DNS configured and working
- DHCP scope created and clients get IP
- File share created with proper permissions
- Users, Groups and OUs created
- Group Policy created and applied
- Clients joined to domain and working
- Server monitored and maintained

14. VERIFICATION CHECKLIST

- Server IP and DNS working
- Can ping SRV-DC01
- Can resolve company.com
- Clients get IP from DHCP
- Can login with domain user
- Can access \\SRV-DC01\CompanyData
- GPO settings applied on client
- No critical errors in Event Viewer

15. USEFUL COMMANDS

Command	Purpose
ipconfig /all	View IP configuration
nbtstat -a company.com	Test DNS resolution
nltest /r	Check Domain Controller health
gpupdate /force	Force Group Policy update
net user	List domain users
net share	List shared resources
servermanager	Open Server Manager
windows start backup /backupTarget:C:	Start system backup

16. KEY FILES AND TOOLS

Tool / File	Purpose
Server Manager	Manage roles and features
Active Directory Users and Computers	Manage AD objects
DNS Manager	Manage DNS zones
DHCP Manager	Manage DHCP scopes
Group Policy Management	Create and manage GPOs
Event Viewer	View server logs
Windows Server Backup	Backup and restore

KEY POINTS

- AD DS is the core of Windows Server environment.
- DNS is required for domain functionality.
- DHCP automates IP address assignment.
- File sharing must have proper NTFS and Share permissions.
- Group Policy ensures consistent configuration.
- Regular monitoring and backup are essential.
- Always follow least privilege security principle.

NOTE: This lab is for educational purposes. In production, follow organizational policies, security best practices, and perform proper backups before making changes.

